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A PROJECT REPORT

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"AI FOR RURAL INNOVATION AND SUSTAINABLE SYSTEMS"

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BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

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This is to certify that the Project Report titled **"AI FOR RURAL INNOVATION AND SUSTAINABLE SYSTEMS"** is a bonafide work carried out by **Shreya A Hurakadli(USN 1RG23CS066), Sneha K V(USN 1RG23CS067), Spoorthi M (USN 1RG22CS070) and Tanushree S R (USN 1RG23CS061)** in partial fulfillment for the award of **Bachelor of Engineering** in **Computer Science and Engineering** of the **Visvesvaraya Technological University, Belgavi**, during the year **2025-2026.** It is certified that all corrections/suggestions given for Internal Assessment have been incorporated in the report. This project report has been approved as it satisfies the academic requirements in respect of project work (BCS685) prescribed for the said degree.

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DECLARATION

We hereby declare that the project work entitled **"AI FOR RURAL INNOVATION AND SUSTAINABLE SYSTEMS"** submitted to the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**, is record of an original work done by us under the guidance of **Dr. Arudra A,** Head of the Department and Associate Professor, Department of Computer Science and Engineering, RGIT, Bengaluru in the partial fulfillment of requirements for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering.** The results embodied in this project have not been submitted to any other University or Institute for award of any degree or diploma.

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ABSTRACT

This project focuses on leveraging Artificial Intelligence (AI) to drive rural innovation and promote sustainable development. Rural communities often face challenges such as limited access to information, inefficient agricultural practices, poor market connectivity, and lack of resource management systems. The proposed solution integrates AI-based tools to address these issues by enabling data-driven decision-making and improving overall productivity.

The system utilizes data collected from remote sensors, satellite imagery, and local inputs to provide real-time insights on soil health, crop conditions, weather patterns, and resource usage. Machine learning models analyze this data to recommend optimal farming practices, predict crop yields, and detect potential risks such as pest infestations or climate impacts. Additionally, the platform facilitates better market access by connecting farmers with buyers and providing price forecasting.

The project also emphasizes sustainability by promoting efficient use of water, energy, and other natural resources. It incorporates intelligent irrigation systems, climate-aware planning, and resource optimization techniques to minimize environmental impact. Furthermore, user-friendly interfaces, including mobile applications in regional languages, ensure accessibility for rural users. By empowering communities with actionable insights and digital connectivity, the system enhances decision-making, improves livelihoods, and supports long-term ecological balance. Overall, this AI-driven approach contributes to building resilient and self-sufficient rural ecosystems.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Agriculture remains the primary source of livelihood for a large share of the rural population in India, and it continues to be the foundation of rural economies across the developing world. Despite its importance, farming is still carried out under conditions of considerable uncertainty. Rainfall is irregular, soil fertility differs from one plot to the next, pests and diseases can destroy a crop within days, and the price a farmer finally receives depends on market conditions that he or she can rarely see in advance. Every one of these factors is related to the others, yet in practice they are usually examined one at a time and often only after a problem has already appeared in the field.

Artificial Intelligence (AI) and Machine Learning (ML) offer a way of handling this complexity. When soil test values, weather forecasts, satellite vegetation indices, sensor readings, photographs of leaves and market prices are brought together, patterns appear that are difficult for a human observer to notice. These patterns can be converted into simple recommendations: which crop suits a particular soil, when the field should be irrigated, which disease is affecting a leaf and where the harvest should be sold. The project reported in this document, **AI for Rural Innovation and Sustainable Systems**, is an attempt to put these ideas together into one working system that supports a farmer through the complete life cycle of a crop.

The system is implemented as a web-based decision-support platform named **KrishiSense** (AI4RISS Intelligence). It begins with soil testing and crop selection, produces a stage-wise growth and irrigation guide, converts the guide into a 19-step farm execution plan that runs from Day 1 to Day 95, monitors pests and diseases with computer vision, tracks field conditions with sensor data and finally connects the farmer to nearby mandis (APMC market yards) through price comparison and least-cost freight routing. The prototype has been calibrated for Kolar district of Karnataka, where tomato, chilli, ragi, maize and groundnut are widely cultivated on red sandy loam soils.

1.2 Problem Statement

Rural farming communities face problems of low productivity, shifting weather, inefficient use of water and fertilizers, and limited access to timely technical information. Data about soil, weather, crop health and markets exist, but it is spread across different sources and formats, and it is rarely combined in a way that a farmer can use in the field. Existing digital tools generally concentrate on a single task, such as a weather application, a price-reporting service or a crop-disease classifier, so the farmer must move between several applications and interpret the results alone.

There is therefore a need for a single, easy-to-use system that accepts farm and environmental data, processes it with AI and ML models, and returns clear advice for every stage of cultivation, from soil assessment to selling the produce. The system must work for users with limited technical education, must be available in regional languages, and should respect the privacy of the farm data that it learns from.

1.3 Motivation

The motivation for this work comes from the gap between what technology can do and what a small farmer receives. A smallholder who plants tomato in September must decide on the variety, the quantity of nitrogen, phosphorus and potassium to apply, the number of litres of water per plant, the days on which to spray, and the market to which the fruit should be sent. Mistakes at any of these points reduce profit. Over-irrigation wastes water and encourages blight; delayed pest detection increases pesticide use; selling in the wrong market can reduce income by a significant margin.

Three observations motivated the design of the project. First, advice is most valuable when it is specific to the local soil, climate and crop calendar rather than generic. Second, farmers need a sequence of actions with dates, because a crop is a process in time and not a single decision. Third, sustainability must be built into the recommendations, which means saving water, reducing chemical inputs and cutting transport losses, so that higher yield does not come at the cost of the environment.

1.4 Importance of AI in Sustainable Agriculture

Sustainable agriculture aims to meet present food needs without damaging the resources that future harvests depend on. AI contributes to this aim in several ways. Machine learning models estimate the suitability of a crop for given soil and weather conditions and thereby avoid

unsuitable plantings. Computer-vision models identify diseases at an early stage, which allows targeted treatment instead of blanket spraying. Optimization algorithms schedule irrigation according to crop stage and forecast rainfall, which reduces water consumption. Graph algorithms find the cheapest route between farm and market, which reduces fuel use and spoilage.

Recent reviews of the field [1], [3], [6] show that AI-driven agriculture is moving from isolated prediction tasks towards integrated socio-technical systems [7]. The same reviews also warn that technology which is not understandable, affordable and accessible will not be adopted by small farms [2], [8]. These two lessons, integration and usability, guide the whole design of the present project.

1.5 Background of the Study

Digital agriculture has developed through several overlapping stages. The first stage introduced weather stations and remote sensing, which gave farmers and planners access to rainfall and vegetation data. The second stage added low-cost Internet of Things (IoT) sensors that measure soil moisture, temperature and humidity inside the field. The third stage applied machine learning to these data to predict yield, classify disease and recommend crops. The most recent stage focuses on integration, explainability and community learning, in which many farms contribute to a shared model without revealing their private data.

In Karnataka, Kolar district is a representative semi-arid region. It is classified within the Eastern Dry Zone, receives low and erratic rainfall, depends heavily on borewell irrigation and is known for its large tomato market at the Kolar APMC yard. Because water is scarce and tomato prices fluctuate sharply, the district provides a realistic test bed for a system that combines crop planning, water saving and market access. The knowledge base of the prototype therefore uses Kolar soil nutrient distributions, local cash crops and nearby mandis.

1.6 Need for the Study

Several needs justify the study. A farmer needs one platform instead of many separate applications. A farmer needs advice that follows the crop calendar and can be checked off step by step. A farmer needs a diagnosis tool that works on a mobile phone camera and gives results with a confidence value. Cooperative bodies and extension workers need a way to learn from many farms without collecting private data. Finally, a system that is meant for rural users must be usable in the user's own language and, where reading is difficult, through speech.

The literature reviewed in Chapter 2 confirms that integration of sensor, weather, satellite and historical data into one decision-support framework is still missing in most existing work. This project addresses that missing link by building and demonstrating such a framework.

1.7 Aim and Objectives of the Project

The aim of the project is to design and develop an AI-driven decision-support system that guides a farmer through the entire crop life cycle, from soil testing to harvesting and market access, while promoting efficient and sustainable use of resources.

The specific objectives are:

- To collect and integrate soil, weather, sensor, satellite-index and market information into one coherent data flow.
- To build an on-device crop recommendation model that ranks crops by suitability for the entered soil nutrients, pH, temperature, humidity, rainfall and planting date.
- To generate crop-specific growth stages, water budgets and a dated irrigation calendar for the selected crop.
- To convert the recommendation into a guided, trackable 19-step farm execution plan covering preparation, growth, protection and harvest phases.
- To detect leaf diseases and pests from photographs using a lightweight vision model and to link the result to treatment advice.
- To present field conditions and sensor values with adaptive imputation for missing readings.
- To connect farmers with mandi prices, buyers and least-cost transport routes.
- To provide multilingual voice support and a privacy-preserving community model based on federated averaging.

1.8 Scope and Sustainability

The present scope of the project covers a single-district prototype for Kolar, Karnataka, with seven preset crops (tomato, chilli, spinach, ragi, maize, groundnut and pulses) and nine functional modules accessible from one dashboard. The system accepts manually entered soil test values and sensor-style inputs, and it retrieves live weather data through public application programming interfaces. The architecture is deliberately modular, so that new districts, crops, languages and data feeds can be added without changing the overall design.

The sustainability goals of the project are expressed through four indicators shown on the dashboard: expected yield improvement against the regional seasonal baseline, water conserved by the smart drip schedule, pest-health index based on the diagnostic confidence of the vision model, and logistics cost saved by shortest-path routing. Future scope includes integration with real IoT hardware, additional languages, offline operation and extension to other districts, as described in Chapter 8.

1.9 Existing System

At present, a farmer who wishes to obtain digital assistance has to rely on a number of unconnected sources. Weather information is available from meteorological applications. Market prices are published on government portals and in newspapers. Crop advice is obtained from the local Krishi Vigyan Kendra (KVK), from agricultural officers or from fertiliser dealers. Disease identification is mostly visual and depends on personal experience. Where AI tools exist, they are usually research prototypes or commercial applications that address one task only.

The main limitations of the existing arrangement are summarised below.

- Fragmentation: weather, soil, disease and market information are available in different places and formats.
- Generic advice: recommendations are not tied to the local soil nutrient levels or the Kolar crop calendar.
- No continuity: advice is given as a one-time answer and not as a dated sequence of actions that can be tracked.
- Poor accessibility: most tools assume literacy in English and good internet connectivity.
- Privacy concerns: centralised data collection discourages farmers from sharing records.
- Limited market linkage: price comparison and transport planning are seldom connected to crop planning.

1.10 Proposed System

The proposed system, KrishiSense, integrates the required functions into a single dashboard. A farmer opens the Overview page, which displays the day's priority and four sustainability indicators, together with shortcuts to the Crop Advisor, the pest scanner and the voice assistant. The farmer can select a Kolar crop preset or enter soil test values, and the on-device Gaussian Naive Bayes model, trained on 2,200 records, returns a ranked list of crops with suitability

percentages. After a crop is chosen, the system generates a crop guide, an irrigation and water-budget calendar and a 19-step execution plan with dates calculated from the planting date.

During cultivation, the Leaf Disease and IoT Lab accepts a photograph and returns the disease name, confidence, percentage of leaf damaged and treatment suggestions, while the Soil and Sensors page shows live field conditions. Near harvest, the Mandi and Supply Chain module lists modal prices and buyer contacts for the Kolar mandis and finds the least-cost route using Dijkstra's algorithm. A multilingual voice advisor and a federated community model complete the platform. The functional modules and the underlying techniques are listed in Table 1.1.

Table 1.1: Functional modules of KrishiSense and the techniques they use

Module	Purpose	Technique / data source
Live Weather (IMD)	Current weather and 7-day rainfall ensemble	REST API integration with IMD and Open-Meteo
Crop Advisor (AI)	Ranks crops by suitability and creates a crop plan	Gaussian Naive Bayes, 2,200 records, on-device
Leaf Disease and IoT Lab	Diagnoses leaf disease and pests from photographs	MobileNetV3 + YOLOv8 edge model
Soil and Sensors	Shows soil moisture, pH, NDVI and imputed data	ST-GNN sensor fusion, Sentinel-2, ERA5, SoilGrids, NASA POWER
Resource Optimizer	Schedules water and inputs	Multi-agent reinforcement learning (MARL) drip schedule
Mandi and Supply Chain	Mandi prices, buyers and freight route	GCN buyer-seller pricing, Dijkstra shortest path
Community FedAvg	Shared learning without sharing raw data	Federated averaging with differential privacy
Voice Assistant	Spoken advice in regional languages	Speech-to-text and text-to-speech engine

1.11 Outcome of the Project

1.11.1 Technical and Functional Outcomes

A fully functional single-page web application was produced that runs in an ordinary browser, requires no server-side installation for its core crop-advisory model and presents nine integrated modules. The crop model runs locally on the user's device, which allows advice to

be generated even when the connection is weak. A 19-step plan with dates, phase filters, check-boxes and quick-action links was produced for each recommended crop.

1.11.2 Analytical and Research Insights

The project shows that a Naive Bayes classifier is a practical choice for on-device crop recommendation because it is fast, needs little memory and returns probabilities that can be shown to the farmer as suitability scores. It also shows that separating the model inputs from the farm's actual soil-test values, and using the latter only to personalise alerts and the 90-day plan, keeps the recommendation transparent.

1.11.3 Educational and Skills Development Outcomes

The team gained practical experience in data preprocessing, machine learning, computer vision, graph algorithms, federated learning, user-interface design, and technical writing. The project also led to a research paper titled "AI for Rural Innovation and Sustainable Systems" published in the International Journal of Scientific Research in Engineering and Management (IJSREM), Volume 10, Issue 10, October 2026.

1.11.4 System Performance Evaluation

The sustainability indicators displayed by the prototype are a +33.5 % yield outlook against the regional seasonal baseline, 72 % water conserved by the smart drip schedule, a 93 % pest-health index based on the confidence of the edge diagnostic model and a 15 % logistics cost saving through shortest-path routing. The community model reports 92.6 % accuracy over 284 edge clients after 12 converged federated rounds at a privacy budget of $\varepsilon = 2.1$. These values are produced by the prototype and are discussed with their limitations in Chapter 6.

1.12 Alignment with Sustainable Development Goals

The project supports several of the United Nations Sustainable Development Goals (SDGs). It contributes to **SDG 2 (Zero Hunger)** by helping farmers choose suitable crops, protect them from losses and raise productivity. It contributes to **SDG 6 (Clean Water and Sanitation)** by scheduling drip irrigation according to the crop stage and the forecast so that less water is used. It contributes to **SDG 9 (Industry, Innovation and Infrastructure)** by bringing intelligent tools to rural areas through an affordable web platform. It contributes to **SDG 12 (Responsible Consumption and Production)** by reducing excess fertiliser and pesticide use and by lowering

post-harvest and transport losses. Finally, it supports **SDG 13 (Climate Action)** because planning from forecasts and early warnings helps farmers adapt to changing weather.

1.13 Report Organization

Chapter 1- Introduction: This chapter introduces the project AI for Rural Innovation and Sustainable Systems. It explains the background of using Artificial Intelligence for rural development and smart agriculture. It presents the need for improving agricultural productivity, reducing resource wastage, and supporting data-driven decision-making. The chapter includes the problem statement, objectives, scope, and an overview of the proposed system.

Chapter 2- Literature Survey: This chapter presents the review of existing research related to Artificial Intelligence, smart agriculture, sustainable rural development, crop prediction, pest and disease detection, and resource optimization. It discusses the approaches used in previous studies and identifies their advantages and limitations. The research gap that motivates the proposed integrated AI-based system is also presented.

Chapter 3 – System Analysis and Requirement Analysis: This chapter analyzes the existing and proposed systems. It explains the limitations of traditional agricultural practices, such as manual decision-making, inefficient resource utilization, lack of timely information, and limited prediction capabilities. It then explains how the proposed AI-based system addresses these limitations through intelligent data analysis and decision support.

Chapter 4 – System Design and Methodology: System design defines the architecture, components, interfaces, and data flow of a system, while methodology describes the systematic approach and steps followed to develop, implement, and evaluate it.

Chapter 5 – Implementation: Implementation is the process of converting the system design into a working solution by developing, integrating, testing, and deploying the required software components according to specified requirements.

Chapter 6 – Testing: Testing is the process of evaluating a system to identify errors, verify that all components function correctly, and ensure the system meets specified requirements and expected performance standards.

Chapter 7 – Snapshots: Snapshots are captured images of the system's interface or important stages, providing visual evidence of implementation, functionality, and results for documentation and demonstration.

Chapter 8 – Conclusion: The system successfully meets its objectives by providing an effective and reliable solution**.**
Future Work: Future improvements can enhance accuracy, scalability, automation, features, and real-world usability.

Chapter 9 – References: References include research papers, books, websites, datasets, documentation, and other reliable sources consulted during the project to understand concepts, develop the system, validate results, and support implementation**.**

1.14 Summary

This chapter introduced the problem of fragmented, generic and inaccessible digital support for rural farmers, explained the motivation and need for the study, stated the aim and objectives, described the existing and proposed systems and outlined the expected outcomes. The next chapter reviews the research on which the project is based.

CHAPTER 2

LITERATURE SURVEY

A literature survey was carried out on the use of Artificial Intelligence, machine learning, the Internet of Things and computer vision in sustainable agriculture, with attention to crop advisory, field monitoring, disease diagnosis, interpretability, rural adoption and privacy-preserving learning. Nine studies were reviewed in detail. For each study, the table gives its name, year, authors, advantages and disadvantages, and the notes that follow explain what the study contributes and how the present project, KrishiSense, builds on it.

2.1 Leveraging AI for Sustainable Agriculture: Interpretable Results

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Leveraging AI for sustainable agriculture: A comprehensive review of recent advances (Artificial Intelligence Review)	2026	A. Rajbongshi et al. [2]	- Comprehensive review of recent AI advances for sustainable agriculture. - Stresses that AI results must be interpretable and understandable to farmers. - Covers monitoring of crops, disease spotting and yield decisions.	- Review article; no working decision-support system is presented. - Does not turn predictions into a dated sequence of field actions. - No link between crop decisions and market access.

This review surveys recent advances in AI for sustainable agriculture and draws attention to the way in which results are presented to the user:

- Machine learning and deep learning are used for crop monitoring, disease spotting and yield decisions.
- Farmers need results that they can understand, not only predictions from complex models.
- Interpretable outputs are presented as a condition for trust and adoption in the field.
- The review identifies a need to connect model output with plain, actionable advice.

The study confirms that explainable output is necessary for farm adoption. It remains a survey and does not offer an integrated tool. KrishiSense applies the lesson by showing crop suitability

as a percentage, disease results with a confidence value and every plan step with plain-language guidance.

2.2 AI-Driven Technologies for Sustainable Crop Production

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Artificial intelligence for sustainable agriculture: A comprehensive review of AI-driven technologies in crop production (Sustainability)	2025	Z. Ali et al. [6]	- Reviews machine learning and deep learning for analysing complex agricultural data. - Shows the potential of AI for sustainable crop production. - Useful basis for choosing model families.	- Models work only when data are available, good and well integrated. - Tasks are treated one at a time. - No end-to-end workflow from soil test to market.

This review examines AI-driven technologies used in crop production:

- Machine learning and deep learning are applied to complex agricultural data.
- Typical tasks are crop selection, yield prediction, soil analysis and water management.
- The usefulness of a model depends on the availability and quality of the data supplied to it.
- The review supports the use of AI to improve productivity while using resources efficiently.

The review shows that crop recommendation from soil and climate data is an established application, but that each task is usually studied alone. KrishiSense keeps the proven idea, a lightweight Gaussian Naive Bayes model on soil nutrients, pH and climate, and extends it into growth guidance, irrigation, protection and harvest in one workflow.

2.3 Digitalization of Agriculture for Sustainable Crop Production

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Digitalization of agriculture for sustainable crop production: A use-case review (Frontiers in Environmental Science)	2024	R. R. Shamshiri et al. [13]	- Reviews digital farming use cases for sustainable crop production. - Highlights wireless sensors, IoT, computer vision and automated decision tools. - Shows that digital farming lets farmers monitor and manage better while using inputs efficiently.	- Use-case review; no common platform is proposed. - Sensor data are not combined with market and planning tools. - Assumes infrastructure that small farms may lack.

This review describes how digital technologies support crop production:

- Wireless sensor networks and IoT checks collect information on soil, crops and weather.
- Computer vision and automated decision tools turn that information into action.
- Digital monitoring helps farmers manage inputs more efficiently.
- The review treats digitalisation as a route to sustainable production.

The work establishes the value of sensing and automated decision tools. It does not connect field data with crop planning and selling. KrishiSense joins sensor tiles, satellite and weather data with the execution plan and the mandi module, and it fills gaps in sensor data by imputation.

2.4 Artificial Intelligence of Things (AIoT) for Smart Agriculture

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Artificial Intelligence of Things (AIoT) for smart agriculture: A review of architectures, technologies and solutions (Journal of Network and Computer Applications)	2024	D. Muhammed et al. [14]	- Reviews architectures, technologies and solutions for AIoT in agriculture. - Gives a reference for structuring a layered smart-farming system. - Combines the strengths of IoT sensing and AI analysis.	- Architecture-level review; no crop-cycle advice is delivered. - Practical issues such as missing data and weak connectivity remain. - No market or community-learning component.

This review studies how AI and IoT are combined for smart agriculture:

- IoT devices gather data from fields, and AI analyses the data to support decisions.
- Different architectures and technology choices are compared.
- Deployment issues such as connectivity and data quality are discussed.
- The review serves as a guide for structuring sensing, processing and application stages.

KrishiSense follows the same staged idea of data collection, processing, AI analysis and user interaction. To reduce the connectivity and data-gap problems noted in this line of work, the core crop model runs on the device and missing sensor values are imputed.

2.5 Machine Learning and Computer Vision in Precision Agriculture

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Systematic review on machine learning and computer vision in precision agriculture: Applications, trends, and emerging techniques (Engineering Applications of Artificial Intelligence)	2025	Y. N. Kuan et al. [12]	- Systematic review of machine learning and computer vision applications. - Covers disease detection and crop monitoring from images. - Identifies trends and emerging techniques.	- Review of techniques; no deployed advisory tool. - Diagnosis alone does not tell the farmer what to do next. - Image-based methods depend on image quality and computing resources.

This systematic review covers the use of images and machine learning in precision agriculture:

- Computer vision is applied to crop monitoring and disease detection.
- Convolutional networks and real-time detectors are the main techniques.
- Trends and emerging techniques are summarised for researchers.
- The practical need is for models that are light enough for field use.

The review supports the choice of image-based diagnosis. KrishiSense uses a lightweight MobileNetV3 classifier with a YOLOv8 detector on the edge, and it links the diagnosis (disease, confidence and percentage of leaf damaged) to treatment advice and to the protection steps of the execution plan.

2.6 Artificial Intelligence in Sustainable Agriculture: A Socio-Technical Roadmap

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Artificial intelligence in sustainable agriculture: Towards a socio-technical roadmap (Smart Agricultural Technology)	2025	P. Wu and Y. Zhong [7]	- Argues for integrated, practical AI systems in place of isolated tools. - Considers technological, environmental and user-related factors together. - Judges AI by lasting benefit to users with different technology levels.	- Conceptual roadmap; no implemented platform is reported. - No concrete workflow for a single crop cycle. - Leaves the design of the user interface to later work.

This paper proposes a move from stand-alone AI tools to socio-technical systems:

- Technology must fit real-world needs such as people, ease of use, scale and adoption.
- The value of agricultural AI should not be judged only by model performance.
- Practical and lasting help to users in different environments is the measure of success.
- Integrated systems are preferred over single-purpose tools.

KrishiSense is built on this view. It is one platform of nine modules, calibrated for a single district, with a checklist-style plan, language selection and voice access so that farmers with different skill levels can use the same system.

2.7 Artificial Intelligence in Agribusiness: Prospects and Barriers

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Leveraging artificial intelligence in agribusiness: A structured review of strategic management practices and future prospects (Discover Sustainability)	2025	I. A. Bhat et al. [8]	- Reviews the growing role of AI in farm businesses. - Lists uses in farming, resource management, real-time checks and data-driven decisions. - Identifies barriers for small farms and developing areas.	- Identifies barriers but does not implement a solution for smallholders. - Weak infrastructure and hard access remain unresolved. - No specific guidance for fitting analysis into everyday farm work.

This structured review examines how AI is used in agribusiness and what holds it back:

- AI supports farming operations, resource management and real-time monitoring.
- Data-driven decisions are named as the main benefit.
- Hurdles for small farms include limited data, weak technology infrastructure and hard access.
- Fitting advanced analysis into everyday farm work is a challenge.

The hurdles listed in this review are the design constraints of KrishiSense. The core model works in a browser without a server, the interface follows the sequence of the farmer's daily tasks, and the step-by-step plan resembles the notebook that many farmers already keep.

2.8 Federated Averaging: Learning from Decentralised Data

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
Communication-efficient learning of deep networks from decentralized data (AISTATS)	2017	H. B. McMahan et al. [15]	- Trains a shared model while data stay on the clients' own devices. - Reduces the number of communication rounds needed. - Works with unbalanced data spread unevenly across clients.	- Model updates alone do not give a formal privacy guarantee. - Not designed for agriculture or for low-connectivity rural settings. - Needs additional protection such as differential privacy [16].

This paper introduced federated averaging (FedAvg), the method used for community learning in KrishiSense:

- Each client trains a model locally on its own data.
- Only the model updates are sent to a server, which averages them in proportion to the number of local samples.
- The averaged model is sent back to the clients for the next round.
- Raw data never leave the client device.

For farm data, this makes learning from many farms possible without collecting private soil, yield or income records. KrishiSense adds clipping and Gaussian noise to the updates, as in differential privacy [16], and displays the privacy budget (epsilon = 2.1) together with the accuracy of the community model.

2.9 YOLO: Real-Time Object Detection

NAME	YEAR	AUTHOR	ADVANTAGES	DISADVANTAGES
You only look once: Unified, real-time object detection (IEEE CVPR)	2016	J. Redmon et al. [18]	- Single-pass detector that predicts boxes and classes together. - Fast enough for real-time use. - Output of box, class and score is easy to read.	- Original benchmarks use natural images, not leaf images. - Small or faint lesions are harder to detect. - Needs domain-specific training data and threshold tuning.

This paper describes the detector family on which the lesion localisation of KrishiSense is based:

- The image is processed in a single forward pass.
- The network predicts bounding boxes and class probabilities together.
- Lightweight variants, and lightweight backbones such as MobileNetV3 [19], suit phones and edge devices.
- Non-maximum suppression removes duplicate boxes.

In KrishiSense, a YOLOv8 detector localises lesion clusters on a leaf photograph, and the area of the boxes gives the percentage of leaf damaged. A box with a label and a confidence value (for example Early blight, 93.0 %) can be read directly by the farmer.

2.10 Literature Survey Summary

The reviewed literature shows that AI and machine learning can support crop selection, yield estimation, soil analysis and water management [6], that digital sensing and IoT make continuous field monitoring possible [13], [14], and that computer vision can detect crop disease from images [12], [18]. The reviews also agree that results must be interpretable and usable by farmers [2], that integrated, socio-technical systems are preferable to isolated tools [7], and that small farms face barriers of data, infrastructure and access [8]. Federated averaging [15] makes it possible to learn from many farms without moving their data.

However, the existing studies mostly address one task or one data source at a time, deliver advice as a one-time prediction, and do not connect the agronomic decision with the transport and selling decision. The following gaps were identified.

- No unified framework combines sensor data, weather information, remote-sensing indices and historical records into one decision-support workflow that covers the entire crop cycle.
- Recommendations are usually delivered as one-time predictions and not as a dated, trackable sequence of field operations that includes protection, harvest and market stages.
- Market access, transport planning and community learning are treated separately from agronomy, so the farmer does not see the link between crop decisions and final income.

KrishiSense addresses these gaps by combining an on-device Gaussian Naive Bayes crop advisor, a growth and irrigation guide, a 19-step execution plan, edge pest and disease diagnosis, sensor monitoring, mandi matching with least-cost routing, multilingual voice access and a privacy-preserving community model in one locally calibrated platform. The requirements derived from this survey are analysed in the next chapter.

CHAPTER 3

SYSTEM ANALYSIS AND REQUIREMENT SPECIFICATION

3.1 System Analysis

System analysis is the study of the users, the data and the processes involved, carried out before design begins. For this project, the analysis considered who will use the system, what information they need at each stage of cultivation, which data sources can supply that information and what constraints the rural environment imposes.

3.1.1 Stakeholders and Users

The primary user is the smallholder farmer who cultivates one to five acres and makes decisions about crop, inputs and sale. Secondary users are agricultural extension officers and Krishi Vigyan Kendra staff, who may use the system to advise many farmers, farmer producer organisations (FPOs), who aggregate produce, and APMC dealers, who are the buyers. Each group needs a different view of the same data, which is why the dashboard is divided into task-specific modules.

3.1.2 Information Needs Across the Crop Cycle

The analysis divided the crop cycle into four phases, each with its own questions. In the preparation phase, the farmer asks which crop suits the soil and how the land should be prepared. In the growth phase the questions concern irrigation, fertiliser and weeding. In the protection phase the farmer needs early warning and treatment for pests and diseases. In the harvest phase the concern is harvest timing, grading, transport and the best market. The 19 steps of the farm execution plan were derived from these questions and are described in Chapter 4.

3.1.3 Data Sources Identified

Soil nutrient values (nitrogen, phosphorus, potassium and pH) are taken from soil health cards or laboratory tests. Weather data is retrieved from Open-Meteo and the India Meteorological Department (IMD). Satellite vegetation index (NDVI) is derived from Sentinel-2, soil temperature from ERA5-Land, solar radiation from NASA POWER and soil classification from SoilGrids. Sensor readings represent soil moisture, ambient temperature and humidity.

Mandi prices and buyer information are taken from APMC records for Kolar and neighbouring taluks. Leaf photographs are supplied by the farmer through the device camera or gallery.

3.1.4 Use-Case Descriptions

The use-case diagram in Fig 4.3 shows nine functions available to the farmer. Table 3.1 describes each use case with its actor, the main flow and the outcome. The weather and data services act as external actors for the first and sixth use cases, and the APMC dealer is linked to the seventh.

Table 3.1: Description of the use cases

ID	Use case	Main flow	Outcome
UC1	Check live weather	Farmer searches a place or uses location; the system calls the weather service and shows the current values and 7-day forecast	Temperature, humidity, wind, rain probability and forecast shown; rainfall shared with the Crop Advisor
UC2	Get crop recommendation	Farmer sets N, P, K, pH, temperature, humidity, rainfall and planting date, then presses Generate	Top three crops with suitability percentage; best crop labelled Top AI Pick
UC3	View growth and irrigation plan	Farmer opens the guide of the chosen crop	Five growth stages, water strategy and a dated irrigation calendar
UC4	Track execution checklist	Farmer marks each of the 19 steps done, in progress or pending and adds field notes	Progress ring shows the percentage completed
UC5	Scan leaf for disease	Farmer selects crop and stage and uploads a leaf photograph	Disease name, confidence, leaf damage, lesion clusters and treatment advice
UC6	View soil and sensor data	Farmer opens the Soil and Sensors page	Ten tiles with values; missing readings filled by imputation

ID	Use case	Main flow	Outcome
UC7	Find mandi price and route	Farmer searches a crop or filters by mandi and runs the freight optimiser	Modal prices, buyer leads and least-cost route
UC8	Ask voice assistant	Farmer selects a language and speaks or taps a prompt chip	Short answer spoken aloud and shown on screen
UC9	Join community model	Farm client trains locally and sends a clipped, noised update	Shared model accuracy, clients, rounds and privacy budget displayed

3.2 Requirement Specification

3.2.1 Functional Requirements

The functional requirements describe what the system must do. They are listed module-wise in Table 3.2.

Table 3.2: Functional requirements

ID	Module	Requirement
FR1	Weather	Retrieve current weather and a 7-day forecast for any searched place or the detected location, and refresh on demand.
FR2	Crop Advisor	Accept N, P, K, pH, temperature, humidity, rainfall and planting date; return ranked crops with suitability percentage.
FR3	Crop Guide	Generate growth stages, soil context, water strategy and a dated irrigation and water-budget calendar for the chosen crop.
FR4	Execution Plan	Produce a 19-step plan with dates, phase filters, progress status (done, in progress, pending), notes and reset option.

ID	Module	Requirement
FR5	Leaf Disease Lab	Accept a leaf image for a selected crop and growth stage; return disease name, confidence, damage percentage and treatment.
FR6	Soil and Sensors	Display soil moisture, temperature, humidity, pH, NDVI, soil temperature, solar radiation, soil class and data-gap fill.
FR7	Resource Optimizer	Recommend a water and input schedule that conserves water while meeting crop demand.
FR8	Mandi and Supply Chain	Search crop prices, filter by mandi, show modal price range and buyer lead; compute the least-cost route.
FR9	Voice Assistant	Provide speech-to-text and text-to-speech advice in selected regional languages.
FR10	Community Model	Show federated model accuracy, number of clients, rounds, privacy budget and activity log.
FR11	Interface	Provide language selection, light and dark mode, online status indicator and crop presets.

3.2.2 Non-Functional Requirements

The non-functional requirements describe the qualities that the system must possess and are given in Table 3.3.

Table 3.3: Non-functional requirements

Attribute	Requirement
Usability	Simple layout, large touch targets, icons with text labels, language selection and voice interaction for users with limited literacy.

Attribute	Requirement
Performance	Crop recommendation should be computed in the browser within a second; weather refresh should complete within a few seconds on a normal mobile connection.
Reliability	The core crop model must work without internet; failures of external APIs must not stop other modules.
Scalability	New crops, regions, languages and data feeds should be added by extending configuration and data, not by redesigning the system.
Privacy and security	Raw farm data stays on the device; community learning exchanges only noisy model updates.
Portability	Runs in common browsers on laptops, tablets and smartphones.
Maintainability	Modular code with separated data, model and interface layers.
Accessibility	High-contrast dark mode and spoken output for users who cannot read easily.

3.2.3 Hardware Requirements

Table 3.4: Hardware requirements

Component	Specification
Development computer	Laptop or desktop with minimum 8 GB RAM, multi-core processor, 256 GB storage
User device	Smartphone, tablet or computer with a modern browser, camera and microphone
Field sensors (optional)	Soil moisture probe, temperature and humidity sensor, wireless module
Connectivity	Internet access for weather and market updates; offline mode for the core crop model

Component	Specification
Power supply	Mains or battery supply for the computer; solar or battery for field sensors

3.2.4 Software Requirements

Table 3.5: Software requirements

Component	Specification
Operating system	Windows, macOS or Linux for development; any OS with a modern browser for users
Front end	HTML5, CSS3, JavaScript single-page application
Machine-learning tools	Python, Scikit-learn, TensorFlow and PyTorch for model development [design stack]
Data storage	Browser storage for local state; MongoDB proposed for centralised storage
External services	Open-Meteo, IMD, Sentinel-2, ERA5-Land, NASA POWER, SoilGrids, speech engine
Cloud (planned)	AWS or Google Cloud for large-scale storage and processing
Other tools	Web browser developer tools, Git, image-editing and documentation tools

3.2.5 System Constraints and Assumptions

The following constraints and assumptions apply to the present implementation.

- The prototype is calibrated for Kolar district; use in other regions requires recalibration of the crop model.
- Soil test values are assumed to be entered correctly from a laboratory report or Soil Health Card.
- Weather and market data depend on the availability and accuracy of the public sources.

- The sensor values and community-model figures shown on the dashboard are prototype outputs and have not yet been validated against long-term field trials.
- Diagnostic output is advisory and does not replace the opinion of an agricultural expert in serious outbreaks.

3.2.6 Crop Knowledge Base

The crop knowledge used by the Crop Advisor and the plan generator was organised in a structured form for the seven Kolar presets. Each entry records the typical duration, the main agronomic sensitivity and the main protection measure. Table 3.6 summarises the knowledge shown by the prototype.

Table 3.6: Crop knowledge base for Kolar presets

Crop	Duration	Main agronomic guidance	Protection and notes
Tomato	85–105 days	Flagship Kolar cash crop; sensitive to water stagnation; drip fertigation with calcium, boron and split potassium	Fruit borer; early blight; pheromone traps; keep foliage dry
Chilli	135 days	Major spice and vegetable cash crop; responsive to drip fertigation and calcium nitrate	Thrips, mites and leaf curl; yellow sticky traps and neem oil
Spinach	40 days	Short-duration leafy vegetable suited to regular light irrigation	Downy mildew; avoid leaf wetness
Ragi	Seasonal	Dry-zone staple millet of Kolar; low water demand	Neck blast; resistant variety and timely sowing
Maize	Seasonal	High nitrogen feeder in red soil; split N into basal, knee-high and tasseling stages for 5+ t/ha	Armyworm; scouting and biological control
Groundnut	Seasonal	Legume suited to red sandy loam; fixes nitrogen	Leaf spots; seed treatment
Pulses	Seasonal	Low-input rotation crop that improves soil nitrogen	Pod borer; pheromone traps

Durations and notes for tomato, chilli, spinach and maize are as displayed by the prototype; the remaining entries are general agronomic guidance and should be confirmed with the local KVK before use.

3.3 Feasibility Study

3.3.1 Technical Feasibility

All the technologies required are mature and openly available. The crop model is a small statistical classifier that runs in the browser. Public weather interfaces are free. Lightweight vision models can run on ordinary phones. The team has working knowledge of Python, JavaScript and machine learning, so the project is technically feasible.

3.3.2 Operational Feasibility

The interface follows the daily workflow of a farmer: check the weather, decide the crop, follow the plan, scan the leaf, sell the crop. Language selection and voice interaction reduce dependence on reading skills, and the step-by-step plan with check-boxes resembles the notebook that many farmers already maintain. Hence the system can be adopted with minimal training.

3.3.3 Economic Feasibility

Because the core model runs locally and the data services are free or low cost, the running cost is small. The development cost consists mainly of student effort. For the farmer, the expected benefits are lower water and input use, fewer crop losses and better prices, which far outweigh the cost of a smartphone and a data connection that many farmers already possess.

3.3.4 Ethical and Social Feasibility

The design follows the guidance of Omotayo et al. [9] on responsible AI. The system is advisory, explains its output with probabilities, avoids collecting raw farm data centrally and does not replace human advisers. Local-language support helps include farmers who might otherwise be excluded.

3.3.5 Risk Analysis

Every system that depends on public data, user input and models carries risks. Table 3.7 lists the main risks identified during analysis and testing, together with the measure that the design takes against each one.

Table 3.7: Risks and their mitigation

Risk	Possible effect	Mitigation in the design
Weather service unavailable	Forecast missing in the Weather module and Crop Advisor	Fallback message, last saved values retained, other modules unaffected
Wrong soil values entered	Misleading alerts and plan	Values to be taken from a laboratory report or the Soil Health Card; field values do not change the crop-match score
Poor leaf photograph	Lower diagnostic reliability	Guidance to photograph in good light with the lesion in focus; expert confirmation for severe outbreaks
Outdated mandi prices	Farmer chooses the wrong market	Prices shown as modal values with a range; farmer advised to confirm with the market
Voice recognition errors	Wrong or missing spoken answer	Quick prompt chips send a fixed intent without speaking
Percentages read as guarantees	Over-reliance on the model	Output described as model confidence and as advisory, not a guarantee of yield
Loss of privacy	Farm records exposed	Core models on the device; only clipped, noised updates shared
Use outside Kolar	Poorly matched advice	Prototype calibrated for Kolar; recalibration required for other regions

3.4 Summary

This chapter analysed the users, data and constraints and specified the functional, non-functional, hardware and software requirements. The feasibility study concluded that the project is technically, operationally, economically and socially feasible. The design that satisfies these requirements is presented in Chapter 4.

CHAPTER 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Overview

KrishiSense is designed as a layered, modular system in which data flows from sources through processing and AI models to decision support and finally to the farmer. The design follows the framework proposed in the published paper: data collection, data processing, data integration, AI/ML analysis, decision support, database management and alert notification. The user interface exposes these capabilities through nine sidebar modules grouped under two headings: Core Intelligence (Overview, Live Weather, Crop Advisor, Leaf Disease and IoT Lab, Soil and Sensors) and Sustainability and Trade (Resource Optimizer, Mandi and Supply Chain, Community FedAvg, Voice Assistant).

4.2 Design Goals

- **Integration:** all farm information reaches the farmer through one interface.
- **Locality:** advice is calibrated to Kolar soil, climate, crops and mandis.
- **Continuity:** advice is expressed as a dated sequence of actions across the crop life cycle.
- **Explainability:** output is shown as suitability percentages, confidence values and plain-language guidance.
- **Efficiency:** core models run on the device with low memory and without a server.
- **Privacy:** community learning shares only noisy model updates.
- **Inclusiveness:** multilingual and voice-based access.

4.3 System Architecture

The architecture has five layers, shown in Fig 4.1. The data-source layer gathers weather, soil-test, sensor, satellite and market inputs. The processing and integration layer cleans, normalises, aligns and imputes the data. The AI/ML engine hosts the crop advisor, the leaf-diagnosis model, the sensor-fusion model, the irrigation optimiser and the mandi-matching model. The decision-support layer converts model outputs into the execution plan, calendars and alerts. The user-interface layer delivers the results through the dashboard, voice advisor and community model.

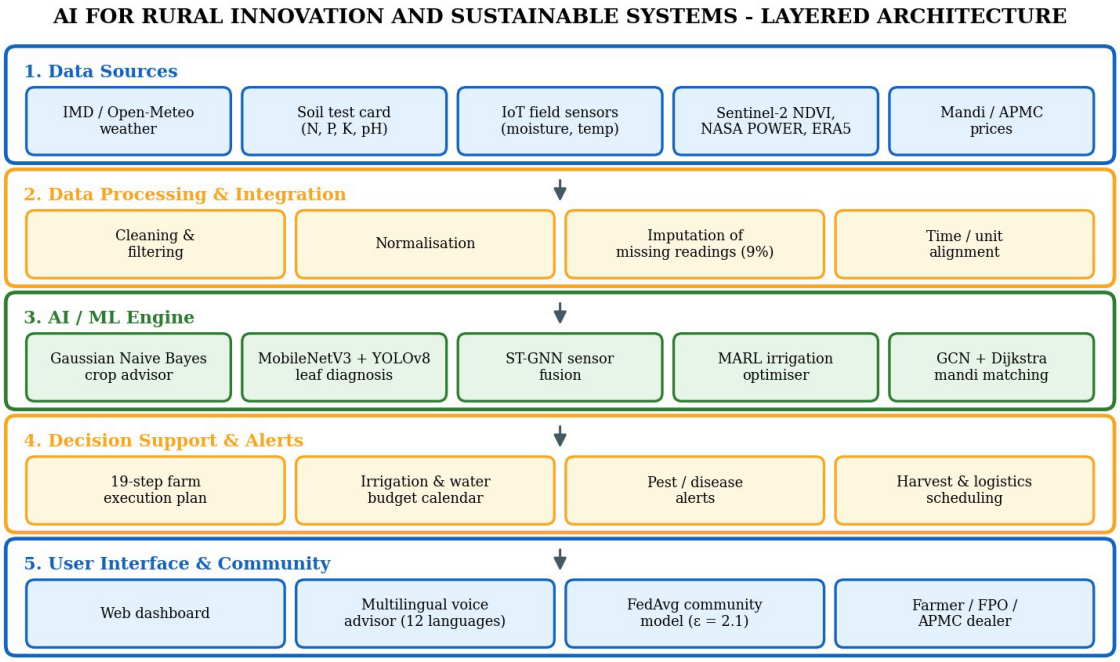


Fig 4.1: Layered architecture of the proposed system

4.3.1 Data Collection and Processing

The Data Collection Module gathers IoT-sensed and remote-sensed information so that field conditions are represented by more than one source. The Data Processing Module prepares this data through cleaning (removal of duplicate and inconsistent records), filtering (removal of noise), normalisation (scaling to a common range) and transformation (encoding for the models). The Data Integration Module aligns formats, time stamps and units so that sensor, weather and satellite values can be analysed together. In the Soil and Sensors page, readings missing from the sensor stream are filled by an adaptive imputation step, and the dashboard reports the share of imputed data (9 % in the test case).

4.3.2 Decision Support, Database and Alerts

The Decision Support Module ranks advice by urgency and presents it in a form that can be acted on, such as which crop to plant, when to irrigate, how much fertiliser to apply and how to manage pests. The Database Management Module stores raw, processed and model-output data and keeps history for trend analysis and model training. The Alert and Notification Module generates alerts for weather warnings, disease risk, irrigation need and market-price changes and gives priority to critical alerts.

4.4 Data Flow and Use Cases

Fig 4.2 shows the level-1 data flow. The farmer supplies soil values, the planting date and leaf images. The weather and satellite services supply live forecasts and indices. Process 1.0 analyses soil and crop suitability, 2.0 generates the growth plan and irrigation schedule, 3.0 diagnoses pests and diseases, 4.0 matches markets and logistics, and 5.0 produces alerts, voice replies and reports. Three data stores hold the soil and field records, the crop and disease models and the mandi price data.

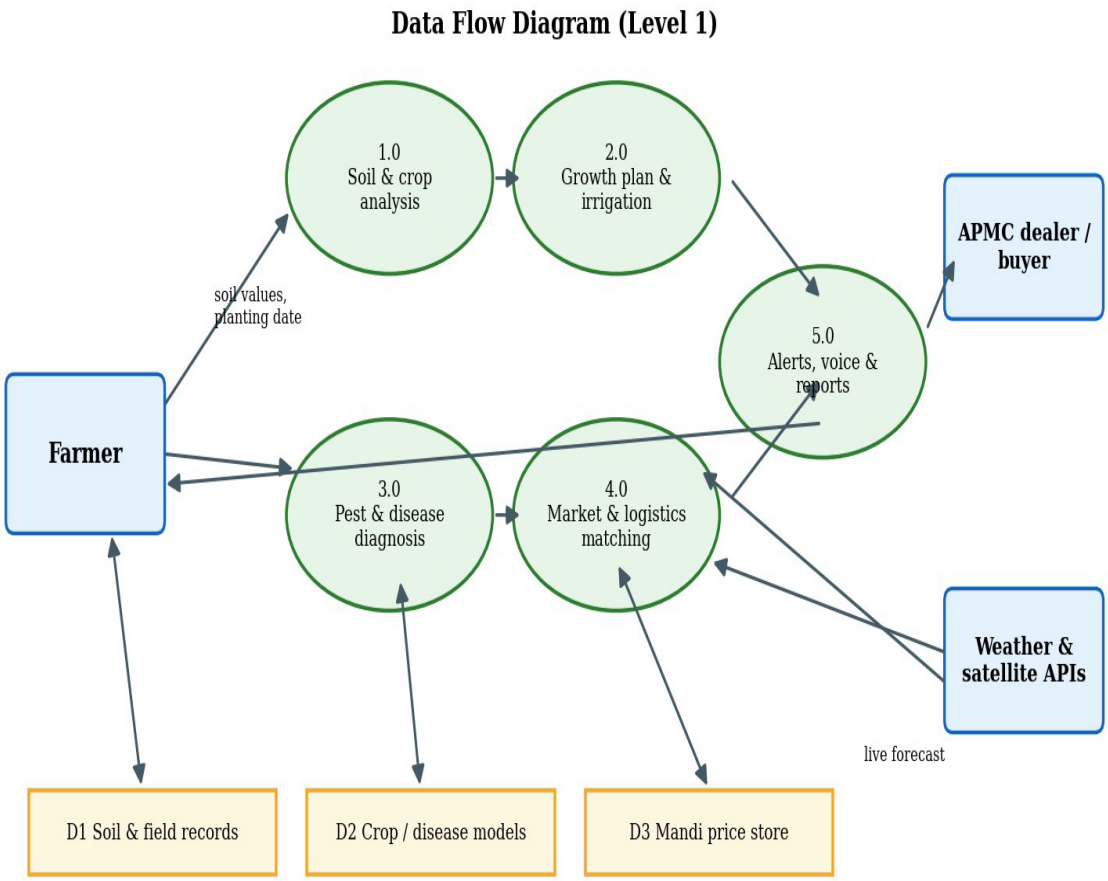


Fig 4.2: Level-1 data flow diagram

The use cases of the system are shown in Fig 4.3. The farmer is the main actor and interacts with all modules. The APMC dealer or buyer is linked to the market module, and the weather and data services are external actors that feed the weather and sensor modules.

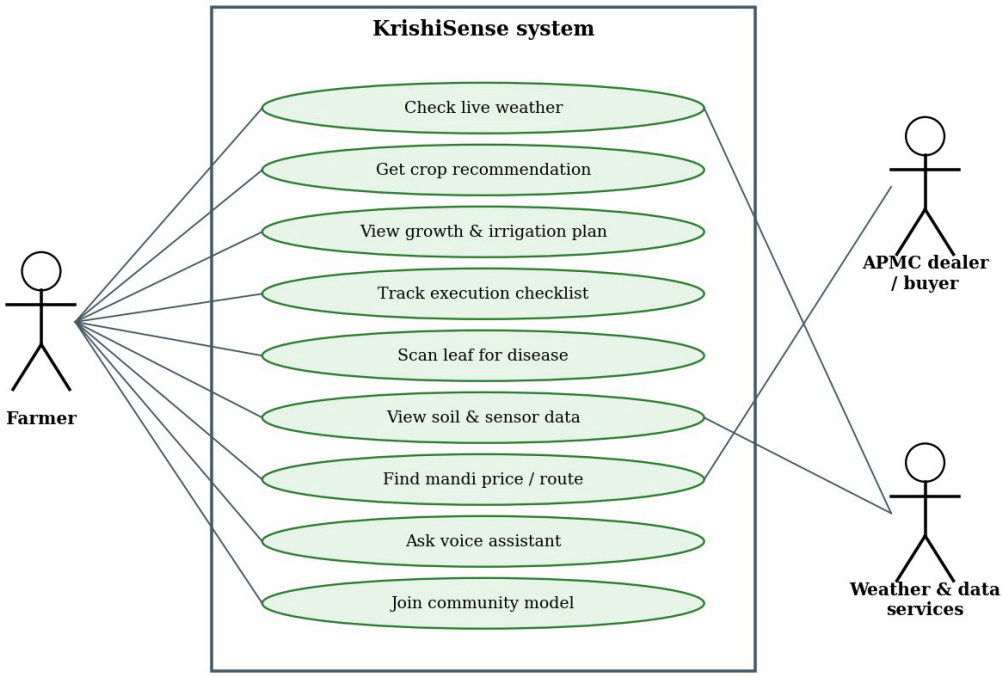


Fig 4.3: Use-case diagram of KrishiSense

4.5 Module Design

4.5.1 Overview Dashboard

The Overview page opens with a greeting, connectivity status, language selector and dark-mode switch. A row of Kolar crop presets (Tomato, Chilli with a 135-day duration, Spinach with 40 days, Ragi, Maize, Groundnut and others) sets typical soil and climate values with one tap. A hero panel shows the priority of the day, "Find the Right Crop and Conserve 35 % Water", with buttons to run the Crop Advisor, scan a pest photograph or open the voice assistant. Four cards display the yield outlook, water conserved, pest-health index and logistics cost saved.

4.5.2 Live Weather Module

The weather module retrieves live data through REST calls to Open-Meteo and IMD and supports place search, quick regions (Kolar, Bengaluru, Ludhiana, Nashik, Guntur) and automatic location detection. It shows temperature, feels-like temperature, humidity, wind, precipitation and rain probability for the base station and a 7-day ensemble of minimum and maximum temperature and rainfall. The forecast is shared with the Crop Advisor so that expected rainfall can be used as an input.

4.5.3 Crop Advisor Based on Gaussian Naive Bayes

The Crop Advisor takes seven numerical inputs: nitrogen (N), phosphorus (P), potassium (K), soil pH, ambient temperature, air humidity and expected seasonal rainfall. A Gaussian Naive Bayes classifier trained on 2,200 records computes for each crop class c the posterior probability

$$P(c \mid x) \propto P(c) \times \prod P(x_i \mid c), \text{ with } P(x_i \mid c) = (1 / \sqrt{(2\pi\sigma^2)}) \times \exp(-(x_i - \mu)^2 / 2\sigma^2),$$

where μ and σ are the mean and standard deviation of feature i in class c estimated from the training data. The class posteriors are normalised and shown as percentages. The model is calibrated for Kolar Zone 5 (Eastern Dry Zone) red sandy loam soils and the regional crops tomato, ragi, maize, groundnut and pulses. Naive Bayes was selected because it trains quickly, needs little memory, handles small datasets well and returns probabilities that are easily explained to the user. The step-by-step flow of the advisor is given in Fig 4.4.

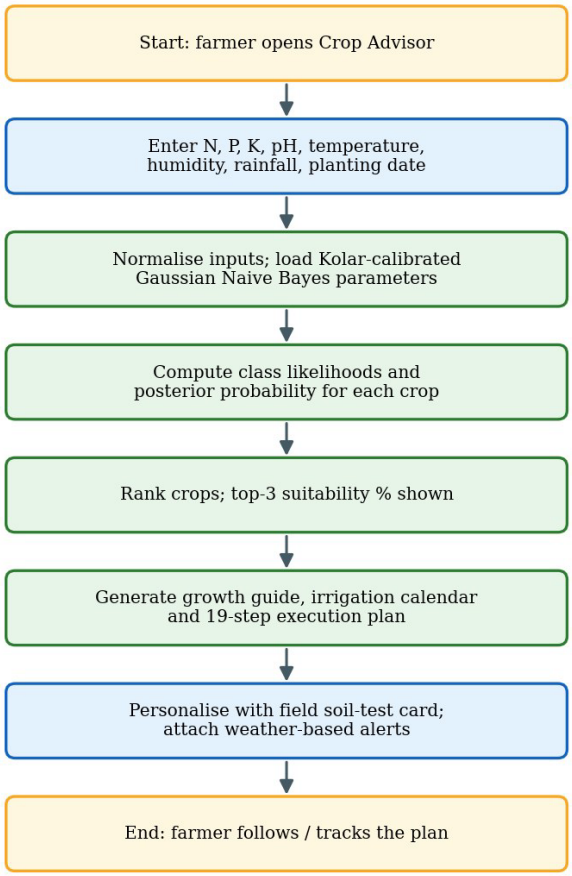


Fig 4.4: Workflow of the Crop Advisor

The page also contains a separate panel, Kolar field soil-test context, in which the farmer enters the available N, P_2O_5 and K_2O in kg/ha and the field pH. These actual field values do not change

the crop-match score; they are used to personalise alerts and the 90-day plan, for example by comparing the field values against reference ranges and advising use of the Soil Health Card for any input rates.

4.5.4 Crop Growth and Resource Guide

After a crop is generated, the guide gives the typical sowing period, the selected sowing date, the duration (85–105 days for tomato) and a water strategy. For tomato the strategy is regular light drip irrigation of 2–3 litres per plant per day while keeping foliage dry to prevent early blight. Five growth stages are described with day ranges: land preparation (1–2 weeks before planting), transplanting (0–15 days), staking and vegetative care (16–35 days), flowering and fruit set (36–60 days) and harvest and grading (61–95 days). An Irrigation and Water Budget Calendar divides the crop period into four intervals with frequency and drip duration, such as every 1–2 days for 45 minutes during establishment, every 2–3 days for 60 minutes in the vegetative stage, every 1–2 days for 75 minutes at flowering and fruit set, and every 3–4 days for 50 minutes during fruit ripening. The calendar notes that, when low seasonal rainfall is entered, the shorter end of each interval should be used and soil moisture checked before irrigating.

4.5.5 Guided 19-Step Farm Execution Plan

The execution plan converts the recommendation into a checklist that runs from Day 1 (1 September 2026) to Day 95 (4 December 2026). It is divided into four phases, as summarised in Table 4.1. Each step shows its day range, a category tag, a title, calendar dates, a description, step guidance and two action links, such as Update Soil Card or Run Irrigation Optimizer and Add Field Note. A three-state control marks each step as done, in progress or pending, and a progress ring shows the percentage completed. The schedule is visualised in Fig 4.5.

Table 4.1: Phases and steps of the farm execution plan

Phase	Steps	Days	Main tools used
1: Preparation (steps 1–5)	Soil assessment; crop and variety selection; land preparation; seed selection and bio-priming; sowing or transplanting	1–15	Soil card, Crop Advisor, soil-moisture check, voice assistant

Phase	Steps	Days	Main tools used
2: Growth (steps 6–9)	Automated irrigation; split fertigation; weed control; NDVI telemetry	16–95	Irrigation optimiser, NPK dosage, sensor dashboard
3: Protection (steps 10–13)	YOLOv8 pest vision; disease early warning; IPM defence; flowering audit	25–86	Pest scanner, disease diagnostics, voice IPM tips
4: Harvest (steps 14–19)	Yield estimation; pre-harvest readiness; harvest; grading and cold care; freight optimisation; mandi access	68–95	Field telemetry, rain forecast, mandi leads, Dijkstra route

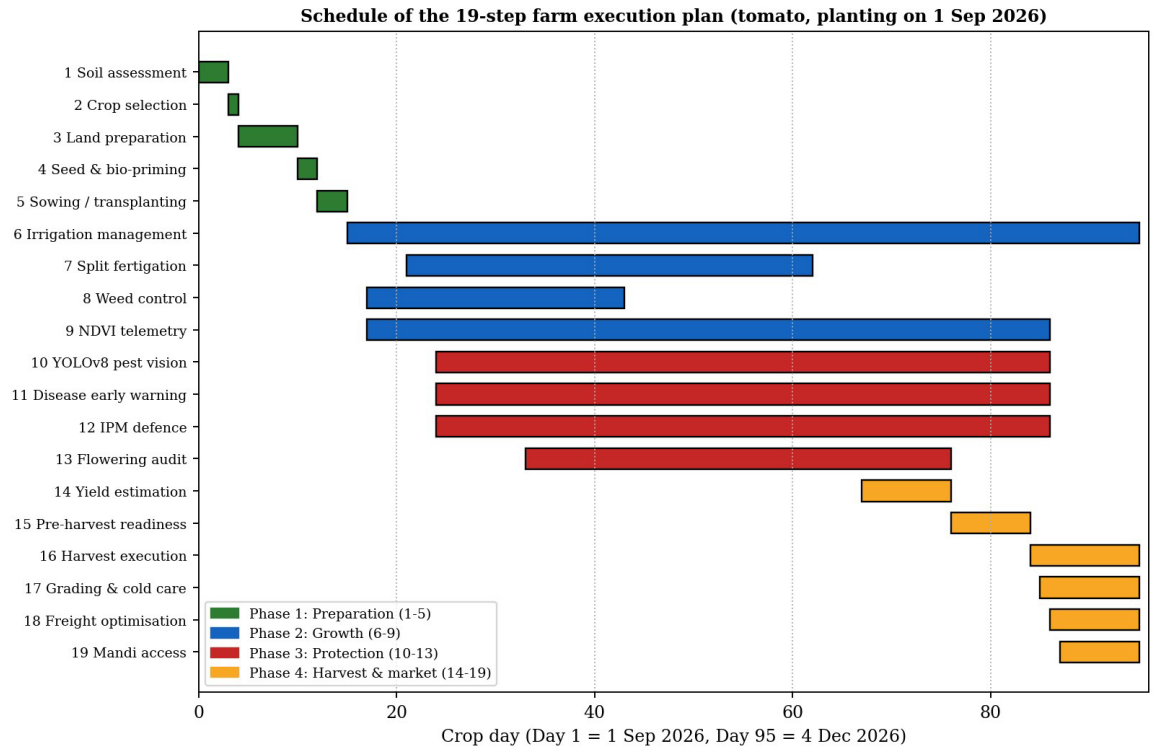


Fig 4.5: Schedule of the 19-step execution plan

4.5.6 Leaf Disease and IoT Lab

The user selects the target crop and the growth stage and drops a farm leaf photograph or chooses a quick sample such as tomato early blight, chilli leaf curl, spinach downy mildew, ragi neck blast or maize armyworm. A lightweight classifier (MobileNetV3) identifies the

disease class and a YOLOv8 detector localises lesion clusters. The output includes the pathogen type, severity band, disease name, confidence, percentage of leaf area damaged and number of lesion clusters, followed by treatment advice. Edge deployment reduces latency and allows use without a continuous internet connection.

4.5.7 Soil and Sensors Module

The Field Conditions and Sensors page shows soil moisture (zone 1), ambient temperature, relative humidity, root-zone pH, NDVI vigour from Sentinel-2, data-gap imputation, soil temperature from ERA5-Land, solar radiation from NASA POWER and the soil class from SoilGrids. A spatio-temporal graph neural network (ST-GNN) fuses readings from several points in space and time and fills missing values, so that the dashboard remains complete even when 15 % of sensor data is missing.

4.5.8 Resource Optimizer

The Resource Optimizer produces a smart drip schedule using multi-agent reinforcement learning (MARL), in which separate agents represent irrigation zones and cooperate to meet crop demand with the least water. The reward is higher when soil moisture is kept within the crop's target band and lower when water is applied that is later lost to drainage or evaporation. The water conserved against a fixed-interval schedule is reported as the sustainability indicator (72 % in the test case).

4.5.9 Mandi and Supply Chain Module

This module lists, for each mandi, the crop, modal price with minimum–maximum range and a buyer lead with contact. Users can search a crop or filter by mandi. The module treats farms, villages, mandis and markets as nodes of a weighted graph whose edge weights represent transport cost. A graph convolutional network estimates buyer-seller prices from the neighbourhood of each node, and Dijkstra's shortest-path algorithm computes the route of least cost, as illustrated in Fig 4.6.

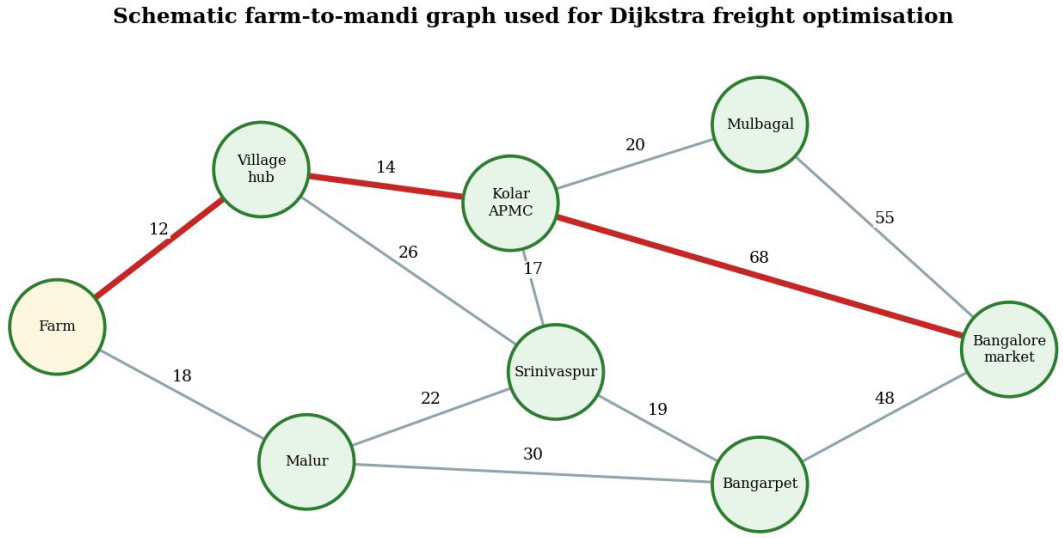


Fig 4.6: Schematic farm-to-mandi graph and least-cost route

Dijkstra's algorithm maintains a tentative distance d(v) for each node, initially infinity for all nodes except the source. At every step, the unvisited node u with the smallest d(u) is fixed, and for each neighbour v the distance is relaxed: if d(u) + w(u, v) < d(v), then d(v) is replaced by this value. The process continues until the destination is fixed. For a graph with E edges and V nodes using a priority queue, the running time is O((V + E) log V), which is trivial for a district-level network.

4.5.10 Multilingual Voice Advisor

The voice advisor provides speech-to-text and text-to-speech across 12 Indian regional languages. The user selects a language, presses Speak Now and asks a question, or taps a quick prompt such as crop suitability status, pest alert and remedy, weather and rain forecast, mandi commodity prices or irrigation advice. The system finds the relevant module, composes a short answer and reads it aloud, so that users who cannot read easily still receive the advice.

4.5.11 Community Model with Federated Averaging

In the community module, each farm client trains a local model on its own device and sends only the model update to an aggregator. With n_k local samples on client k and $n = \Sigma n_k$, the aggregator forms the global weights as $w = \Sigma (n_k / n) w_k$ (federated averaging [15]). Before being sent, each update is clipped and Gaussian noise is added to provide differential privacy [16]. The dashboard shows the accuracy of the global model (92.6 %), the number of clients

(284), the federated rounds (12, converged) and the privacy budget (ε = 2.1), together with an accuracy curve and a live activity log. The process is illustrated in Fig 4.7.

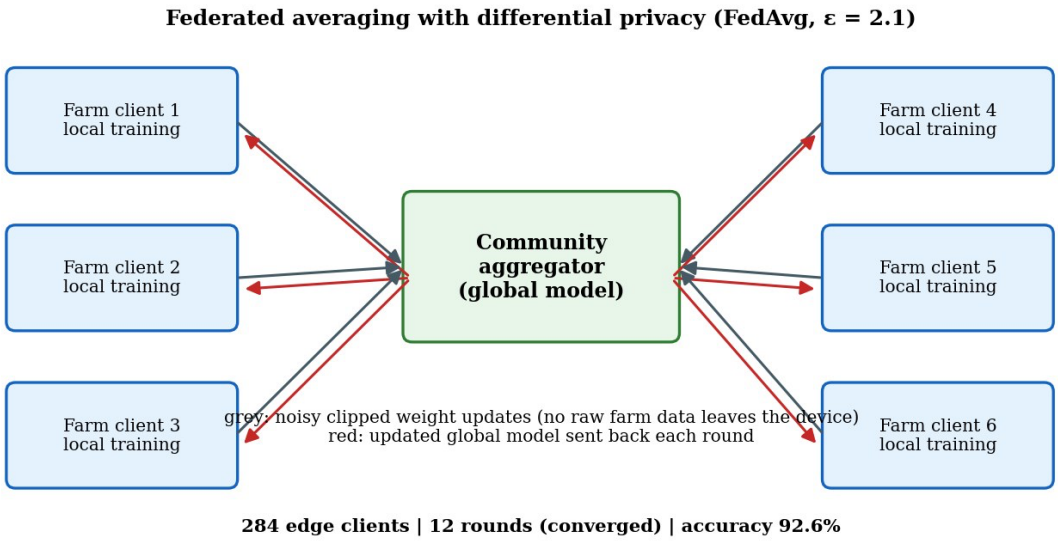


Fig 4.7: Federated averaging with differential privacy

4.5.12 Algorithm for Crop Scoring

The Crop Advisor of Section 4.5.3 can be written as the following procedure. Logarithms are used throughout, and the largest log score is subtracted before exponentiation so that no probability underflows when seven small likelihoods are combined.

Algorithm 4.1: Gaussian Naive Bayes crop scoring

```
Input : x = [N, P, K, pH, T, H, R]; for each crop c: prior(c),
        mean mu(c,i) and standard deviation sd(c,i) of feature i
Output: ranked list of (crop, suitability %)
Where : ln N(v; mu, sd) = -ln(sd*sqrt(2*pi)) - (v-mu)^2 / (2*sd^2)


1. for each crop c:
2.     score(c) = ln prior(c)
3.     for i = 1 to 7:
4.         score(c) += ln N(x[i]; mu(c,i), sd(c,i))
5. m = max over c of score(c)
6. for each crop c: e(c) = exp(score(c) - m)
7. z = sum over c of e(c)
8. suitability(c) = 100 * e(c) / z
9. return crops sorted by suitability, highest first
```

Worked example. To follow the arithmetic, consider a toy case with two crops, two features (soil pH and seasonal rainfall) and equal priors. The parameters below are chosen only for illustration and are not the parameters of the trained 2,200-record model. The input is pH = 6.4 and rainfall = 350 mm.

Table 4.2: Toy example of Gaussian Naive Bayes scoring

Crop	Assumed pH (mean, sd)	Assumed rainfall (mean, sd)	ln prior	ln L (pH)	ln L (rain)	Total score	Suitability
Tomato	6.5, 0.4	300 mm, 60	-0.6931	-0.0339	-5.3605	-6.0876	99.4 %
Maize	6.0, 0.5	700 mm, 120	-0.6931	-0.5458	-9.9599	-11.1988	0.6 %

The larger score belongs to tomato, so m = -6.0876. Then e(Tomato) = 1 and e(Maize) = exp(-5.1112) = 0.0060, giving z = 1.0060 and suitabilities of 99.4 % and 0.6 %. The example also shows why the prototype can return a value as high as 99 % for tomato: when several features agree, the normalised posterior of the best class approaches one. The percentage must therefore be read as model confidence for the entered conditions, as stated in Section 5.4.2.

4.5.13 Worked Example of the Freight Route

Dijkstra's algorithm of Section 4.5.9 is applied here to the schematic graph of Fig 4.6, whose edge labels are illustrative transport-cost units. Starting from the farm, the unvisited node with the smallest tentative distance is fixed at each step and its neighbours are relaxed. Table 4.3 gives the order in which the nodes are fixed.

Table 4.3: Dijkstra's algorithm on the schematic farm-to-mandi graph

Step	Node fixed	Distance from farm	Reached through
1	Farm	0	Source
2	Village hub	12	Farm (12)
3	Malur	18	Farm (18)
4	Kolar APMC	26	Village hub (12 + 14)
5	Srinivaspur	38	Village hub (12 + 26); routes through Malur (40) and Kolar (43) are dearer
6	Mulbagal	46	Kolar APMC (26 + 20)
7	Bangarpet	48	Malur (18 + 30); route through Srinivaspur (57) is dearer
8	Bangalore market	94	Kolar APMC (26 + 68); routes through Bangarpet (96) and Mulbagal (101) are dearer

The least-cost route to the Bangalore market is therefore Farm, Village hub, Kolar APMC, Bangalore market, with a cost of 12 + 14 + 68 = 94 units, which is the red path drawn in Fig 4.6. The cost to the Kolar APMC alone is 26 units. The module compares such route costs with the modal price offered at each mandi, so that a nearer market with a slightly lower price can be preferred over a distant one when freight absorbs the difference.

4.5.14 Worked Example of Federated Averaging

Section 4.5.11 gives the aggregation rule w = sum of (n_k / n) w_k. Consider three farm clients with 100, 300 and 600 local samples, each holding a model of two weights. Table 4.4 shows the weighted average.

Table 4.4: Toy example of federated averaging

Client	Local samples n_k	Share n_k / n	Weight 1	Weight 2
Client 1	100	0.1	0.20	0.50
Client 2	300	0.3	0.40	0.30
Client 3	600	0.6	0.10	0.70
Global model	n = 1000	1.0	**0.20**	**0.56**

The first global weight is 0.1 x 0.20 + 0.3 x 0.40 + 0.6 x 0.10 = 0.20 and the second is 0.1 x 0.50 + 0.3 x 0.30 + 0.6 x 0.70 = 0.56. Client 3 has six times the data of client 1 and so has six times its influence. In the implementation of Listing 5.5 the clients send clipped weight updates rather than raw weights, and Gaussian noise with a standard deviation of sigma x clip / (number of clients) is added to the aggregate before the global model is updated. Clipping limits how much any single farm can change the model, and the noise hides what remains, which together give the privacy budget of epsilon = 2.1 shown on the dashboard.

4.5.15 Interaction between the Modules

Although the nine modules appear as separate sections of one page, they exchange information in a fixed order that follows the farmer's workflow, and this exchange was verified in the integration tests of Section 6.4.2.

- **Weather to Crop Advisor:** the forecast rainfall from the Weather module is passed to the Crop Advisor so that the seasonal rainfall input matches the live forecast.
- **Crop Advisor to guide and plan:** the crop chosen in the Advisor generates the growth guide, the irrigation calendar and the 19-step plan, with dates counted from the planting date.
- **Plan to other modules:** each step carries action links, for example step 6 opens the Irrigation Optimizer, step 10 the pest scanner and step 18 the freight route.
- **Disease Lab to Community log:** every diagnosis is written to the activity log, together with the weather retrieval events, with a time stamp.
- **Weather to sidebar:** the temperature in the sidebar widget is the same value as in the weather card, so the farmer sees it from every page.

Because the interfaces between modules are limited to these values, a failure in one module, such as an unavailable weather service, leaves the others working, as required by the reliability requirement of Table 3.3.

4.6 Data Design

The main data items handled by the system, with their sources and use, are given in Table 4.5.

Table 4.5: Main data items

Data item	Source	Format / unit	Used by
Soil N, P, K, pH	Soil test / model inputs	kg/ha, pH 0–14	Crop Advisor
Temperature, humidity	Weather API / sensor	°C, %	Advisor, sensors
Rainfall	Forecast / seasonal estimate	mm	Advisor, irrigation
Planting date	Farmer input	Date	Plan generator
Leaf image	Camera / gallery	JPEG / PNG	Disease lab
NDVI, soil temp, radiation	Sentinel-2, ERA5, NASA POWER	Index, °C, MJ	Sensors, telemetry
Mandi price and buyer	APMC records	Rs. per quintal	Mandi module
Plan status and notes	Farmer actions	Done / in progress / pending	Execution plan

4.7 Database Design

The database stores raw inputs, processed data and model outputs so that history can be retrieved for trend analysis and retraining, as described for the Database Management Module in the paper. The logical collections are listed in Table 4.6. In the prototype, the farmer-side state (plan status, notes and entered values) is kept in the browser, and the same structure maps directly to a document database such as MongoDB when a central back end is added.

Table 4.6: Logical collections of the database

Collection	Main fields	Written by
farmers	farmer id, language, village, taluk, preferred crops	Interface
fields	field id, soil N, P_2O_5, K_2O, pH, soil class, area	Soil card form
crop_plans	plan id, crop, planting date, steps with dates, status, notes	Plan generator
sensor_readings	zone, time stamp, moisture, temperature, humidity, imputed flag	Sensor module
diagnoses	image reference, crop, stage, disease, confidence, damage %, time	Disease lab
weather_cache	place, time, current values, 7-day forecast	Weather module
mandi_prices	mandi, crop, modal, minimum, maximum price, buyer lead, date	Market module
fed_rounds	round number, clients, accuracy, privacy budget	Community module

4.8 Alert Rules

The Alert and Notification Module converts model outputs and thresholds into prioritised messages. The principal rules are listed in Table 4.7. The disease-risk rule follows the guidance in the execution plan that relative humidity above 85 % together with temperature spikes triggers mildew, blight or wilt, and the irrigation rule follows the instruction to check soil moisture before irrigating when seasonal rainfall is low.

Table 4.7: Alert rules and priorities

Alert	Trigger	Channel	Priority
Weather warning	High rain probability or heavy rainfall in the 7-day forecast	App, SMS	High
Disease risk	Relative humidity above 85 % or temperature spike during cloudy spells	App, voice	High

Alert	Trigger	Channel	Priority
Irrigation due	Soil moisture below target band for the crop stage	App, voice	Medium
Pest scouting	Scheduled twice-weekly scouting during the protection phase	App	Medium
Harvest window	Pre-harvest readiness with rain forecast in the coming week	App, SMS	High
Price change	Change in modal price at a selected mandi	App, SMS	Low

4.9 Privacy and Security Design

Privacy is addressed at three levels. First, the crop advisor and the plan generator run on the device, so soil values and field notes need not leave it. Second, community learning shares only clipped and noised model updates, which bounds what can be inferred about any single farm; the privacy budget of $\varepsilon = 2.1$ is displayed so that the level of protection is visible. Third, the back end proposed for scaling should use authenticated access, encrypted storage and role-based permissions for farmers, extension officers and dealers. Where buyer contact numbers are shown, they should be those of verified APMC dealers.

4.10 Summary

This chapter described the architecture, data flow, use cases and the design of each module, including the algorithms used for crop recommendation, route optimisation and federated learning. The next chapter explains how the design was implemented.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

This chapter explains how the design of Chapter 4 was converted into a working application. The implementation was done in three parallel streams: the data and model stream, which prepared the datasets and trained the models; the logic stream, which implemented the plan generator, routing and community-learning algorithms; and the interface stream, which built the dashboard. The streams were developed separately and integrated into one application, followed by repeated testing and correction, as recommended in the paper [Section 6.2].

5.2 Implementation Environment

The platform was delivered as a single-page web application (index.html with its style and script files) that runs directly in a modern browser, including from the local file system during development. This choice removes the need for a server for the core functions and allows the Crop Advisor to run entirely on the user's device. Python was the main language for data preparation and model development, with Scikit-learn, TensorFlow and PyTorch as the machine-learning libraries and a MongoDB database layer proposed for centralised storage. Table 5.1 lists the environment.

Table 5.1: Implementation environment

Item	Details
Application type	Single-page responsive web application (HTML, CSS, JavaScript)
Model development	Python with Scikit-learn, TensorFlow and PyTorch
Crop model deployment	Gaussian Naive Bayes parameters loaded and evaluated in the browser
Weather data	Open-Meteo / IMD REST interface, 7-day forecast
Satellite and reanalysis data	Sentinel-2 NDVI, ERA5-Land soil temperature, NASA POWER solar radiation, SoilGrids

Item	Details
Interface features	Language selector, dark mode, responsive layout, online indicator
Planned back end	MongoDB for storage, AWS or Google Cloud for scaling

5.3 Implementation of the Weather Module

When the dashboard loads, the system detects the location through the browser or uses the default Kolar base station. It then sends a request to the Open-Meteo interface for current temperature, apparent temperature, relative humidity, wind speed, precipitation, rain probability and the daily maxima, minima and rainfall for seven days. The weather condition code is converted to a label and an icon, such as overcast or foggy. The sidebar widget shows the same temperature so that the farmer can see the weather from every page. A synchronisation time stamp and a refresh button are provided, and the forecast rainfall is passed to the Crop Advisor so that the seasonal rainfall input stays consistent with the live forecast.

Listing 5.1: Representative logic for the weather request

```
async function loadWeather(lat, lon) {
  const url = "https://api.open-meteo.com/v1/forecast" +
    `?latitude=${lat}&longitude=${lon}` +
    "&current=temperature_2m,relative_humidity_2m,wind_speed_10m" +
    "&daily=temperature_2m_max,temperature_2m_min,precipitation_sum" +
    "&forecast_days=7&timezone=auto";
  try {
    const r = await fetch(url);
    const w = await r.json();
    renderCurrent(w.current);      // temperature, humidity, wind
    renderWeek(w.daily);           // 7-day ensemble cards
    setStatus("Live feed synced");
  } catch (e) {
    setStatus("Offline: showing last saved forecast");
  }
}
```

5.4 Implementation of the Crop Advisor

5.4.1 Dataset and Training

The crop model was trained on 2,200 records whose features are nitrogen, phosphorus, potassium, pH, temperature, humidity and rainfall, with the crop as the label. The records were adjusted to represent Kolar red sandy loam soil nutrient distributions and regional crops. The data was cleaned for duplicates and impossible values, and the feature values were checked against agronomic ranges. For each crop class the mean and standard deviation of each feature and the class prior were computed and stored as a compact parameter table that is loaded by the application, so that no heavy library is needed at run time.

5.4.2 Prediction Logic

At prediction time the log-likelihood of the entered values is computed for every class, the class priors are added, and the results are converted to normalised probabilities. Logarithms are used to avoid numerical underflow when multiplying seven small probabilities. The top three crops are displayed as cards with a suitability bar, a percentage and a short agronomic note, and the best crop is labelled Top AI Pick.

Listing 5.2: Representative Gaussian Naive Bayes scoring

```
function scoreCrops(x, model) {            // x = [N,P,K,pH,T,H,R]
  const logp = {};
  for (const crop in model) {
    let s = Math.log(model[crop].prior);
    x.forEach((v, i) => {
      const {mu, sd} = model[crop].f[i];
      s += -Math.log(sd * Math.sqrt(2 * Math.PI))
           - ((v - mu) ** 2) / (2 * sd * sd);
    });
    logp[crop] = s;
  }
  const m = Math.max(...Object.values(logp));
  let z = 0;
  for (const c in logp) { logp[c] = Math.exp(logp[c] - m); z += logp[c]; }
  return Object.entries(logp)
    .map(([c, p]) => ({crop: c, pct: Math.round(100 * p / z)}))
    .sort((a, b) => b.pct - a.pct);
}
```

Because the normalised posterior is dominated by the best-fitting class when seven features agree, a suitability of 99 % for tomato and a low value for an unsuited crop such as maize is

expected in the Kolar test case. These percentages should be read as model confidence for the entered conditions and not as a guarantee of yield.

5.5 Implementation of the Growth Guide and Execution Plan

Crop knowledge is stored as structured data: for each crop, the typical sowing windows, duration range, water strategy, five growth stages with day ranges, four irrigation intervals with frequency and drip duration, and the 19 steps with their day ranges, categories, descriptions, guidance texts and action links. When the farmer generates a plan, the planting date is read, each day range is converted to calendar dates by adding the day offset to the planting date, and the cards are rendered. For a planting date of 1 September 2026, Day 95 falls on 4 December 2026, which is the end date shown in the plan.

Listing 5.3: Representative conversion of day ranges to dates

```
function addDays(d, n) {
  const t = new Date(d); t.setDate(t.getDate() + n); return t;
}
function buildPlan(plant, steps) {
  return steps.map(s => ({
    ...s,
    start: addDays(plant, s.from - 1),
    end:   addDays(plant, s.to - 1),
    status: loadStatus(s.id) || "pending"   // done | progress | pending
  }));
}
function progress(plan) {
  const done = plan.filter(s => s.status === "done").length;
  return Math.round(100 * done / plan.length);   // e.g. 0 of 19 = 0 %
}
```

The phase filters (All Steps, Phase 1 to Phase 4) show or hide the cards by phase. The three-state control stores the status of each step, the progress ring updates the percentage of steps done, and the reset button clears the checklist. The Add Field Note link lets the farmer attach a short observation to a step, and the action links open the related module, so that, for example, step 6 opens the Irrigation Optimizer and step 18 opens the freight route.

5.6 Implementation of the Disease and Sensor Modules

For disease diagnosis, the uploaded image is resized and normalised, passed through the classifier to obtain the disease class and confidence, and passed through the detector to obtain bounding boxes of lesion clusters. The area of the boxes relative to the detected leaf area gives the percentage of leaf damage, which is converted to a severity band (for example medium for

22.5 % damage). The result is shown beside the annotated image, with labels such as Early 93.0 % on each cluster and a status line identifying the model, crop, growth stage and leaf area. Each diagnosis is also written to the activity log of the community module.

For the sensor module, readings arrive as time series from several zones. Missing values are estimated from neighbouring zones and earlier readings by the graph model, and the percentage of imputed values is reported. Each tile shows the current value with a progress bar that indicates its position within the agronomic range, so that a farmer can see at a glance whether a value is low, normal or high.

5.7 Implementation of the Mandi and Logistics Module

The mandi table is loaded from structured records of APMC yards in Kolar, Srinivaspur, Bangarpet, Mulbagal and Malur. Each record has the mandi name, taluk, a short descriptor such as Asia's second largest tomato hub or major spice hub, the crop, the modal price with range per quintal and the buyer name and phone number. Search and filter controls reduce the table to the crop or mandi of interest. The route finder builds a graph from distances and freight rates and runs Dijkstra's algorithm from the farm node to every candidate mandi, after which the lowest-cost route to each mandi can be compared with the price offered there.

Listing 5.4: Representative Dijkstra shortest-path routine

```
function dijkstra(graph, src) {
  const dist = {}, prev = {}, seen = new Set();
  for (const v in graph) dist[v] = Infinity;
  dist[src] = 0;
  while (seen.size < Object.keys(graph).length) {
    let u = null;
    for (const v in dist)
      if (!seen.has(v) && (u === null || dist[v] < dist[u])) u = v;
    if (u === null || dist[u] === Infinity) break;
    seen.add(u);
    for (const [v, w] of Object.entries(graph[u])) {
      if (dist[u] + w < dist[v]) { dist[v] = dist[u] + w; prev[v] = u; }
    }
  }
  return {dist, prev};
}
```

5.8 Implementation of Voice and Community Modules

The voice module uses a speech recognition and synthesis engine for 12 Indian languages. The recognised text is matched against intents (crop suitability, pest alert, weather, mandi prices

and irrigation), the answer is built from the current state of the relevant module, and it is spoken aloud and shown on screen. The quick prompt chips send a fixed intent without speaking, which is useful in noisy fields.

The community module simulates 284 clients training locally and a server aggregating their updates with differential privacy. In each round the updates are clipped, averaged in proportion to the number of local samples, and perturbed with Gaussian noise before the global model is updated. The accuracy curve of the global model is drawn over the rounds, and events from other modules, such as weather retrieval and leaf diagnosis, are written to the activity log with time stamps.

Listing 5.5: Representative federated averaging with noise

```
function fedAvgRound(globalW, clients, clip, sigma) {
  const n = clients.reduce((a, c) => a + c.n, 0);
  const agg = new Array(globalW.length).fill(0);
  for (const c of clients) {
    const dW = clipNorm(sub(c.localTrain(globalW), globalW), clip);
    for (let i = 0; i < agg.length; i++) agg[i] += (c.n / n) * dW[i];
  }
  for (let i = 0; i < agg.length; i++)
    agg[i] += gaussian(0, sigma * clip / clients.length);
  return add(globalW, agg);                 // new global weights
}
```

5.9 Interface Implementation

The interface uses a dark green sidebar with the KrishiSense logo and two grouped navigation lists, a light content area with cards, and a footer widget with the current place and temperature. Colours are green for positive indicators, amber for caution and red for risk. The layout is responsive, with a four-column indicator row on wide screens that collapses to fewer columns on narrow screens, and a dark mode is provided for use in bright sunlight and at night. All controls have text labels in addition to icons, and the language selector in the header changes the labels and the voice language.

5.10 Data Preparation Steps

The datasets used for the models were prepared with the preprocessing steps described in the paper. They are summarised in Table 5.2 together with the purpose of each step in this project.

Table 5.2: Data preparation steps

Step	Operation	Purpose in this project
Cleaning	Remove duplicate and inconsistent records, impossible values such as pH outside 0–14	Reliable training of the crop model
Filtering	Remove noisy readings from sensor series	Stable sensor tiles and trends
Normalisation	Scale readings to a common range	Fair comparison of features and progress bars
Transformation	Encode crop and disease labels, convert dates to day offsets	Model input and plan generation
Integration	Align units, formats and time stamps of weather, sensor and satellite data	Single coherent field view
Imputation	Estimate missing sensor values from neighbours and history	Complete dashboard despite data gaps

5.11 Page-wise Interface Implementation

Each sidebar entry is implemented as a section of the same page, and navigation scrolls to the section and highlights the active item. The sections and their principal controls are listed in Table 5.3.

Table 5.3: Interface sections and controls

Section	Principal controls and displays
Overview	Greeting, status, language, theme, crop presets, hero actions, four indicator cards
Live Weather	Place search, quick regions, location button, base-station card, 7-day ensemble, refresh

Section	Principal controls and displays
Crop Advisor	Seven sliders, date field, soil-test fields, generate button, crop cards, crop guide, irrigation calendar
Execution Plan	Progress ring, phase filters, 19 cards, status control, action links, field notes, reset
Leaf Disease and IoT Lab	Crop and stage selectors, upload area, sample chips, annotated image, result card
Soil and Sensors	Ten sensor tiles with progress bars and refresh
Mandi and Supply Chain	Crop search, mandi filter, price table, buyer leads, route graph
Voice Assistant	Language selector, Speak Now, Stop, waveform, prompt chips
Community FedAvg	Four metric tiles, accuracy graph, activity log

Error handling was implemented so that a failure in one section does not stop the others. If a weather request fails, a message is displayed and the last values are retained; if the microphone is not permitted, the prompt chips remain usable; and if an uploaded file is not an image, the lab asks for another file.

5.12 Summary

This chapter described the environment, data, algorithms and interface of the implementation. The listings given are representative of the logic used in each module. The next chapter describes the testing carried out on the implemented system.

CHAPTER 6

TESTING

6.1 Introduction

Testing is the process of checking that the implemented system meets its requirements and behaves correctly under normal and abnormal conditions. This chapter explains the purpose of testing, the strategy followed, the test environment, the test cases and the results obtained from the prototype, and it discusses the evaluation metrics proposed for a full-scale evaluation.

6.2 Purpose of Testing

- To verify that each module produces the outputs specified in the functional requirements.
- To confirm that the modules work together, for example that the weather forecast reaches the Crop Advisor and that the plan dates follow the planting date.
- To check usability: layout, labels, language selection and dark mode.
- To check robustness when data is missing, an external service is unavailable or the input is outside the expected range.
- To record the values displayed by the prototype and to identify the limitations that must be addressed before field deployment.

6.3 Testing Environment

Testing was carried out on a laptop with a modern desktop browser, and the responsive layout was checked on a smartphone-sized window. Live weather calls were tested with an internet connection and the fallback behaviour without it. The test location was Kolar, Karnataka, the test crop was tomato, and the planting date was 1 September 2026. Test data included the model inputs N = 92, P = 50, K = 70, pH = 6.4, temperature 27.5 °C, humidity 54 % and seasonal rainfall 20 mm, and the field soil-test values N = 240 kg/ha, P_2O_5 = 45 kg/ha, K_2O = 220 kg/ha and pH = 6.45.

6.4 Testing Strategy

6.4.1 Unit Testing

Each function was tested alone with known inputs: the date-conversion function for step ranges, the Naive Bayes scoring function, the Dijkstra routine on small graphs with known shortest paths, the federated averaging function on toy weights and the formatting functions for prices and dates. Boundary values such as pH 0 and 14, zero rainfall and a leap-year planting date were included.

6.4.2 Integration Testing

Integration tests checked that the output of one module becomes the correct input of the next: the crop selected in the Advisor generates the guide and the plan, the plan action links open the correct module, diagnosis events appear in the community log, and the temperature in the sidebar matches the weather card.

6.4.3 System Testing

The whole workflow was run from a farmer's viewpoint: open the dashboard, check the weather, enter soil values, generate the crop plan, mark steps done, diagnose a leaf, look up mandi prices, ask the voice assistant a question and view the community model.

6.4.4 Usability and Acceptance Testing

The team walked through the complete workflow as a first-time user would, using only the labels on the screen, and noted steps that were unclear. Structured acceptance testing with farmers, extension officers and APMC dealers is planned for the field-validation stage described in Chapter 8.

6.5 Test Cases

The principal test cases and their outcomes are listed in Table 6.1.

Table 6.1: Test cases and results

ID	Test case	Expected result	Observed result
T1	Load dashboard	Overview with four indicators and presets	Displayed correctly

ID	Test case	Expected result	Observed result
T2	Fetch weather for Kolar	Temperature, humidity, wind, rain probability	23 °C, 86 %, 3 km/h, 95 %
T3	7-day forecast	Seven cards with min/max and rainfall	Seven cards, 5 Oct to 11 Oct
T4	Search another place	Weather updates for new place	Updated for chosen place
T5	Crop Advisor with test inputs	Tomato ranked first	Tomato 99 %, Chilli 76 %, Maize 12 %
T6	Enter field soil values	Values appear in guide but score unchanged	Context shown in guide; score unchanged
T7	Generate crop plan	Guide with stages and calendar	Five stages, four irrigation intervals
T8	Plan dates	Day 1 = planting date, Day 95 = +94 days	1 Sep 2026 to 4 Dec 2026
T9	Mark step status	Progress ring updates	Updates; reset clears
T10	Phase filter	Only steps of selected phase shown	Correct step numbers shown
T11	Leaf diagnosis sample	Disease, confidence, damage shown	Early blight, 93.0 %, 22.5 %, 2 clusters
T12	Sensor tiles	All ten tiles with values	Moisture 60 %, pH 6.4, NDVI 0.69, gap 9 %
T13	Mandi search	Rows filtered by crop	Tomato Rs.1,450/q; Chilli Rs.3,400/q
T14	Freight route	Least-cost path highlighted	Path found on test graph
T15	Voice quick prompt	Spoken and written answer	Answer produced for each chip
T16	Community metrics	Accuracy, clients, rounds, epsilon	92.6 %, 284, 12, $\varepsilon = 2.1$

ID	Test case	Expected result	Observed result
T17	Weather service unavailable	Fallback message, other modules work	Message shown; other modules unaffected
T18	Dark mode and language switch	Theme and labels change	Changed correctly

6.5.1 Boundary and Negative Test Cases

In addition to normal cases, boundary and negative inputs were designed to check the robustness of the system. They are listed in Table 6.2.

Table 6.2: Boundary and negative test cases

ID	Input	Expected behaviour	Module
B1	pH at 0 and 14	Model accepts range limits; no error	Crop Advisor
B2	Seasonal rainfall of 0 mm	Shorter irrigation intervals advised; model runs	Advisor, guide
B3	Planting date on 29 February of a leap year	Correct day offsets and end date	Plan generator
B4	Empty search in the mandi table	Full table shown	Mandi module
B5	Non-image file uploaded	Message asks for a valid image	Disease lab
B6	Microphone permission denied	Prompt chips remain usable	Voice module
B7	Place name not found	Message shown; previous weather retained	Weather
B8	Missing sensor values	Imputed value used and gap percentage updated	Sensors

6.6 Results and Observations

6.6.1 Crop Recommendation

For the Kolar test case the Crop Advisor ranked tomato first with a suitability of 99 %, chilli second with 76 % and maize third with only 12 %, as shown in Fig 6.1. The ranking is agronomically sensible: tomato and chilli are the main cash crops of the district and suit red loam with the moderate nitrogen and potassium supplied, whereas maize is a heavy nitrogen feeder and its low match reflects the limited rainfall and humidity entered.

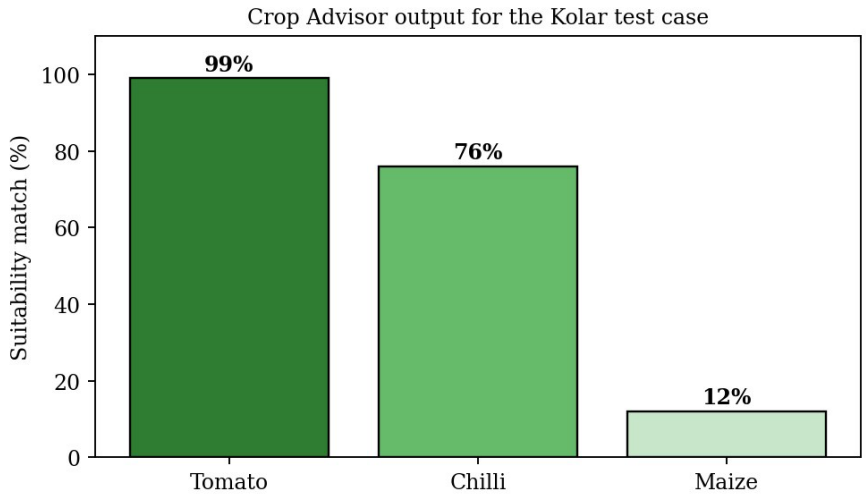


Fig 6.1: Suitability scores returned by the Crop Advisor

6.6.2 Sustainability Indicators

Table 6.3 lists the indicators shown on the dashboard together with the technique that produces each of them. These are prototype outputs: the yield figure compares the model's expected result with the regional seasonal baseline, and the water figure compares the smart drip schedule with a fixed-interval schedule.

Table 6.3: Sustainability indicators displayed by the prototype

Indicator	Value	Basis
Yield outlook	+33.5 %	Improvement against regional seasonal baseline
Water conserved	72 %	Smart MARL drip schedule against fixed schedule
Pest-health index	93 %	Confidence of the YOLO edge diagnostic model

Indicator	Value	Basis
Logistics cost saved	15 %	Dijkstra shortest-path routing
Community model accuracy	92.6 %	FedAvg, 284 clients, 12 rounds, ε = 2.1

6.6.3 Response and Usability

The crop model produced results immediately in the browser, since only seven features and a handful of classes are evaluated. Weather and image-related calls depend on the network and the device. The labels, icons, language selector and quick prompts were designed so that the workflow is understandable without training, but this must be confirmed in trials with farmers.

6.7 Evaluation Metrics for Future Validation

As proposed in the paper, a full evaluation should use measures suited to the task. For regression tasks such as yield prediction, the mean absolute error (MAE), root mean square error (RMSE) and coefficient of determination (R^2) are appropriate. For classification tasks such as crop recommendation and disease diagnosis, accuracy, precision, recall and F1-score should be reported, together with a confusion matrix, using a held-out test set and cross-validation. For optimisation modules the water saved, input saved and transport cost reduced should be measured in actual field trials against farmers' normal practice. For the community model, accuracy should be tracked against the privacy budget.

6.7.1 Proposed Field-Validation Protocol

The values in Table 6.3 are prototype outputs and have not yet been confirmed on farms. Table 6.4 proposes how each module could be validated in the field-validation stage described in Chapter 8, using the evaluation measures of Section 6.7. It is a plan for future work; no results are claimed here.

Table 6.4: Proposed validation of each module

Module	What to measure	Comparison	Measures
Crop Advisor	Agreement of the top crops with the recommendation of an extension officer or KVK and with crops that performed well	Held-out test set and cross-validation	Accuracy, precision, recall, F1-score, confusion matrix
Yield outlook	Recorded yield on participating plots	Farmer's normal practice and the regional seasonal baseline	MAE, RMSE, R-squared
Resource Optimizer	Litres of water per plant per day over the season	Fixed-interval irrigation on a comparison plot	Water saved, input saved
Leaf Disease Lab	Diagnoses of real leaf photographs	Diagnosis confirmed by an agricultural expert	Accuracy, precision, recall, F1-score, confusion matrix
Soil and Sensors	Imputed values for readings deliberately hidden	Real sensor readings	MAE, RMSE
Mandi and logistics	Actual freight cost of loads sent	Cost of the shortest-path route	Transport cost reduced
Community model	Accuracy of the global model at different privacy budgets	Accuracy without noise	Accuracy against epsilon
Usability	Ability of first-time users to finish the workflow without training	Acceptance testing with farmers, extension officers and APMC dealers	Task completion and user feedback

The study would be run over more than one season on farms in several Kolar taluks. A baseline would be recorded before the platform is used, farmers would take part with informed consent,

and raw farm records would remain on their devices, in line with the privacy design of Section 4.9. The results would then be used to recalibrate the crop model and to replace the prototype indicators with measured ones.

6.8 Limitations Identified

- The crop model is trained on a dataset adjusted to Kolar conditions and should be validated with larger field-collected data before wider use.
- Sensor, optimisation and community-model values are produced by the prototype and have not yet been confirmed by multi-season field trials.
- Leaf-disease results depend on image quality and lighting and should be confirmed by an expert for severe outbreaks.
- Mandi prices depend on the freshness of the data source and change daily.
- Voice recognition accuracy varies between languages, dialects and noise conditions.

6.9 Summary

The testing showed that all the planned modules work as specified and work together in a complete workflow from soil test to mandi. The limitations listed above define the direction of the future work described in Chapter 8. The following chapter presents the snapshots of the working system.

CHAPTER 7

SNAPSHOTS

7.1 Overview of System Outputs

This chapter presents the screen outputs of the final working system. The screenshots were captured from KrishiSense running in a browser for the Kolar test case, with the crop tomato and a planting date of 1 September 2026. They are given in the order in which a farmer would use the system, from the first screen to the market and community pages.

7.2 Dashboard Overview

Fig 7.1 shows the Overview page. The left sidebar carries the application name and the nine modules under Core Intelligence and Sustainability and Trade, and the footer shows the current location and temperature (Kolar, 23 °C). The header provides the greeting, an online indicator, the language drop-down and the dark-mode switch. Below it, the strip of Kolar crop presets allows the farmer to set typical conditions for tomato, chilli, spinach, ragi, maize and groundnut quickly. The hero panel announces the priority of the day.

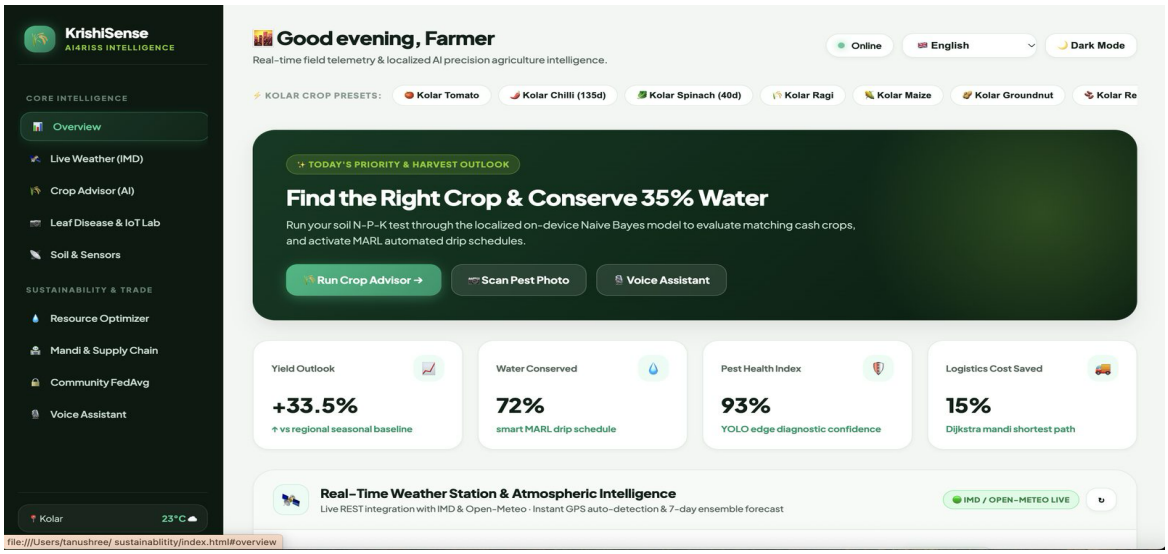


Fig 7.1: Dashboard overview of KrishiSense

7.3 Real-Time Weather and Atmospheric Intelligence

Fig 7.2 shows the weather module. The search box accepts any city, district or village, and quick-region buttons allow one-tap selection of Kolar, Bengaluru, Ludhiana, Nashik or Guntur, together with a location button for automatic detection. The status line confirms that the live

feed has been synchronised. The Kolar base-station card shows 23 °C, overcast and foggy conditions, a feels-like temperature of 27 °C, humidity of 86 %, wind of 3 km/h, precipitation of 0.0 mm and a rain probability of 95 %. The 7-day atmospheric and rainfall ensemble lists the daily minimum and maximum temperature and the rainfall expected, for example 10.1 mm on Tuesday 6 October, and is described as synchronised automatically with the ML Crop Advisor.

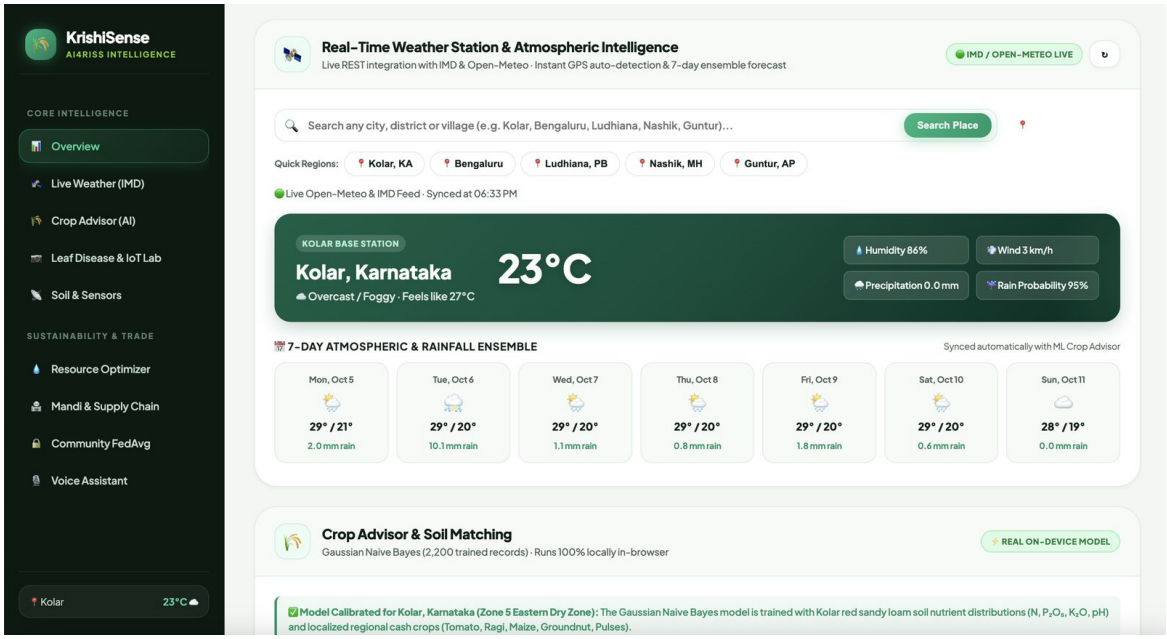


Fig 7.2: Real-time weather station and 7-day rainfall ensemble

7.4 Crop Advisor and Soil Matching

Fig 7.3 shows the Crop Advisor. A notice confirms that the model is calibrated for Kolar, Karnataka (Zone 5, Eastern Dry Zone). Seven sliders set the model nitrogen (92), phosphorus (50), potassium (70), soil pH (6.4), ambient temperature (27.5 °C), air humidity (54 %) and expected seasonal rainfall (20 mm), and a date field holds the planned planting date (01/09/2026). The panel Kolar field soil-test context contains the actual laboratory values (N 240 kg/ha, P_2O_5 45 kg/ha, K_2O 220 kg/ha and pH 6.45) and states that these values personalise alerts and the 90-day plan and do not change the crop-match score. After the button Generate Recommended Crop Plan is pressed, three crop cards appear: Tomato (Top AI Pick) at 99 %, Chilli at 76 % and Maize at 12 %, each with a short agronomic note, for instance that tomato is sensitive to water stagnation and fruit borer and should receive drip fertigation with calcium, boron and split potassium.

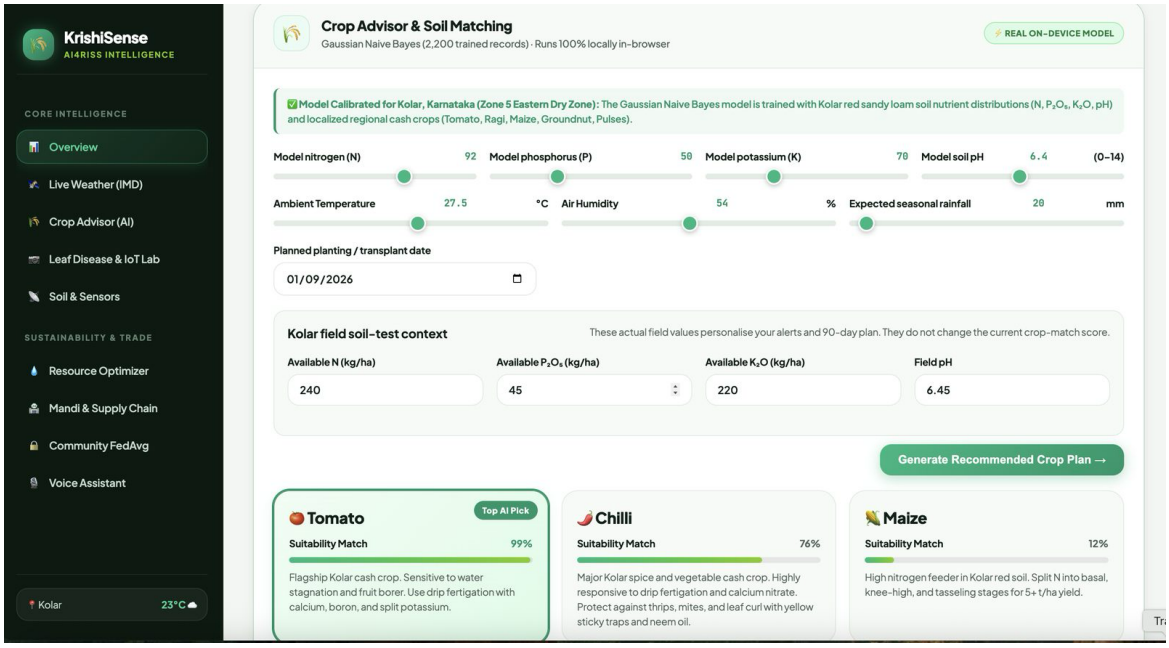


Fig 7.3: Crop Advisor and soil-matching inputs and results

7.5 Crop Growth and Resource Guide

Fig 7.4 shows the personalised Tomato Growth and Resource Guide. The water strategy recommends regular light drip irrigation at 2–3 litres per plant per day while keeping the foliage dry. The soil-test context bar repeats the field values and states that they are within the displayed reference ranges and tha t the Soil Health Card should be used to set input rates.

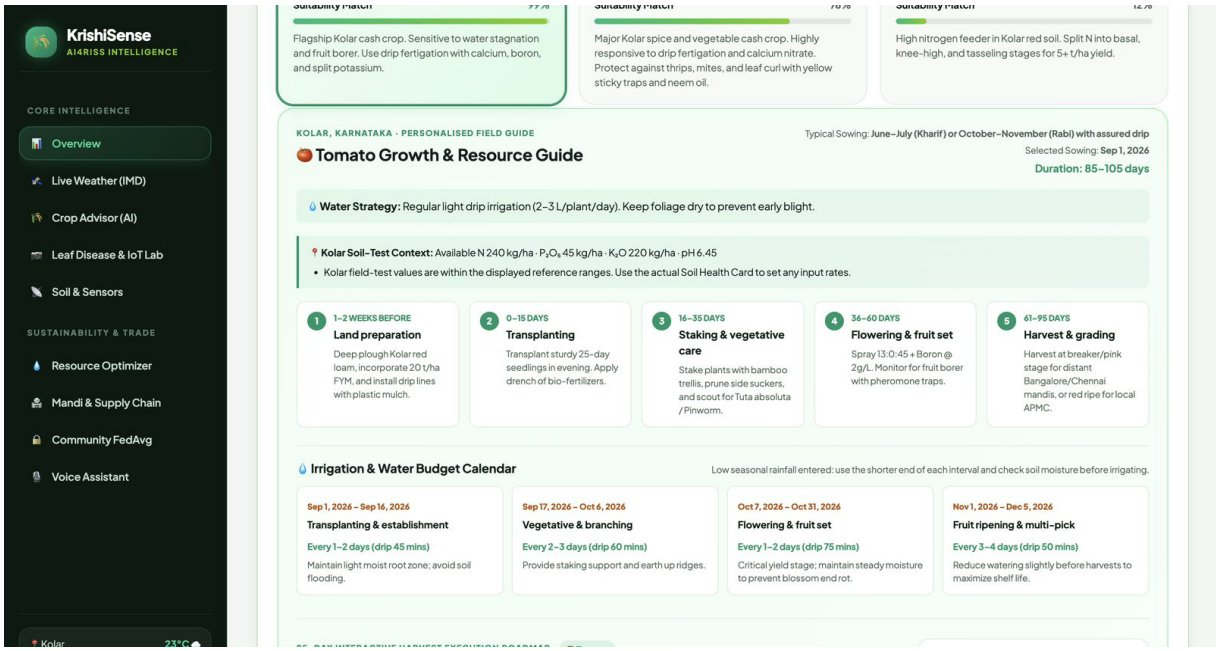


Fig 7.4: Tomato growth stages, soil context and irrigation calendar

7.6 Guided Farm Execution Plan

Fig 7.5 shows the start of the Guided Step-by-Step Farm Execution Plan, which spans Day 1 (1 September 2026) to Day 95 (4 December 2026) and is tracked with a progress ring reading 0 % with 0 of 19 steps done, none in progress and 19 pending.

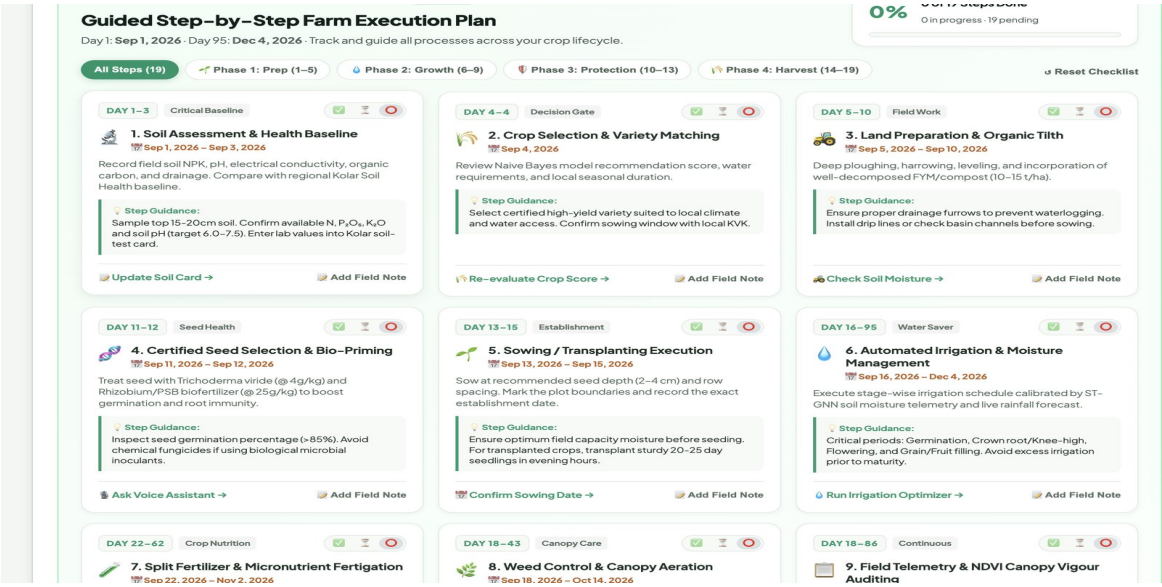


Fig 7.5: Execution plan: phase filters and steps 1 to 6

Fig 7.6 continues with steps 7 to 15: Split Fertilizer and Micronutrient Fertigation, Weed Control and Canopy Aeration, Field Telemetry and NDVI Canopy Vigour Auditing, the YOLOv8 Edge Pest Diagnostic Vision Lab, Disease Early Warning and Micro-Climate Triggers

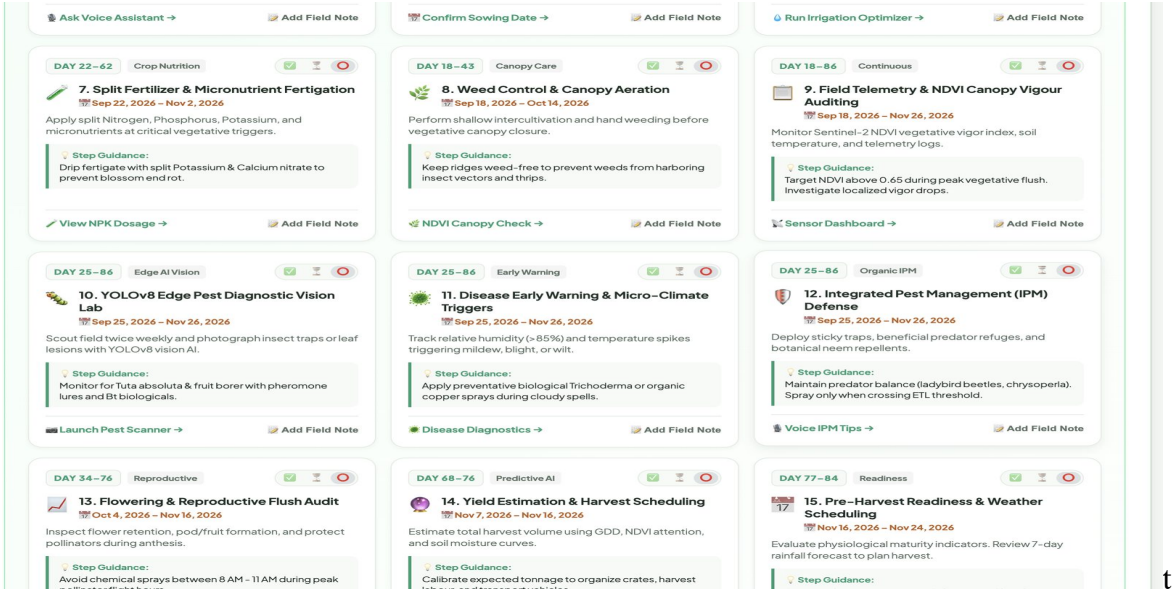


Fig 7.6: Execution plan: steps 7 to 15

Fig 7.7 shows the final steps 16 to 19: Harvest Execution and Field Handling, Post-Harvest Sorting, Grading and Cold Care, Dijkstra Logistics Freight Optimization, and Mandi Market Access and Direct Dealer Connection. These steps lead the farmer from harvest at the breaker or red-ripe stage, through grading (clean grading fetches 15–20 % higher modal price in Kolar APMC yards) and ventilated crate packing, to connecting directly with verified APMC dealers.

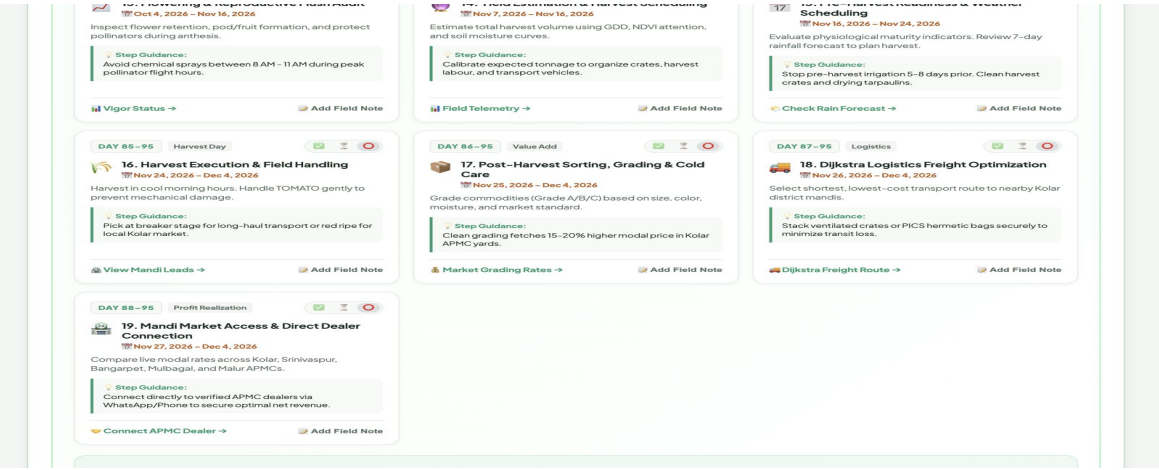


Fig 7.7: Execution plan: steps 16 to 19

7.7 Leaf Disease Diagnosis and Field Sensors

Fig 7.8 shows the AI Crop Stage Leaf Disease and IoT Diagnostic Lab on the left and the Field Conditions and Sensors panel on the right. The sensor panel, labelled ST-GNN fusion with 15 % data gap and adaptive imputation, shows soil moisture 60 %, ambient temperature 27.2 °C, relative humidity 69 %, root-zone pH 6.4, NDVI vigour 0.69, data-gap fill 9 %, soil temperature

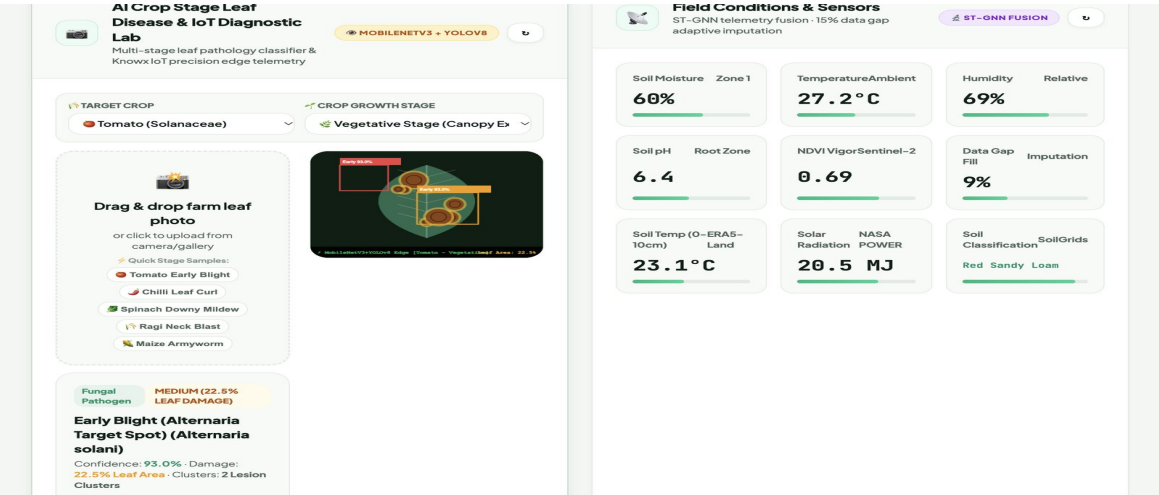


Fig 7.8: Leaf disease and IoT diagnostic lab with field sensor tiles

7.8 Agri-Market Access and Mandi Matching

Fig 7.9 shows the Agri-Market Access and Mandi Matching module, which is labelled GCN with Dijkstra. A search box and a drop-down (All Kolar Mandis) filter the table. The first rows list the APMC Market Yard of Kolar, described as Asia's second-largest tomato hub, with a tomato modal price of Rs. 1,450 per quintal (range Rs. 1,200–1,750) and a chilli modal price of Rs. 3,400 per quintal (range Rs. 2,900–4,100), along with buyer leads and contact numbers. Further rows list the other mandis of the district. The price table lets the farmer compare markets before harvest, and the module also contains a farmer-to-mandi route graph, part of which is visible at the top of Fig 7.10.

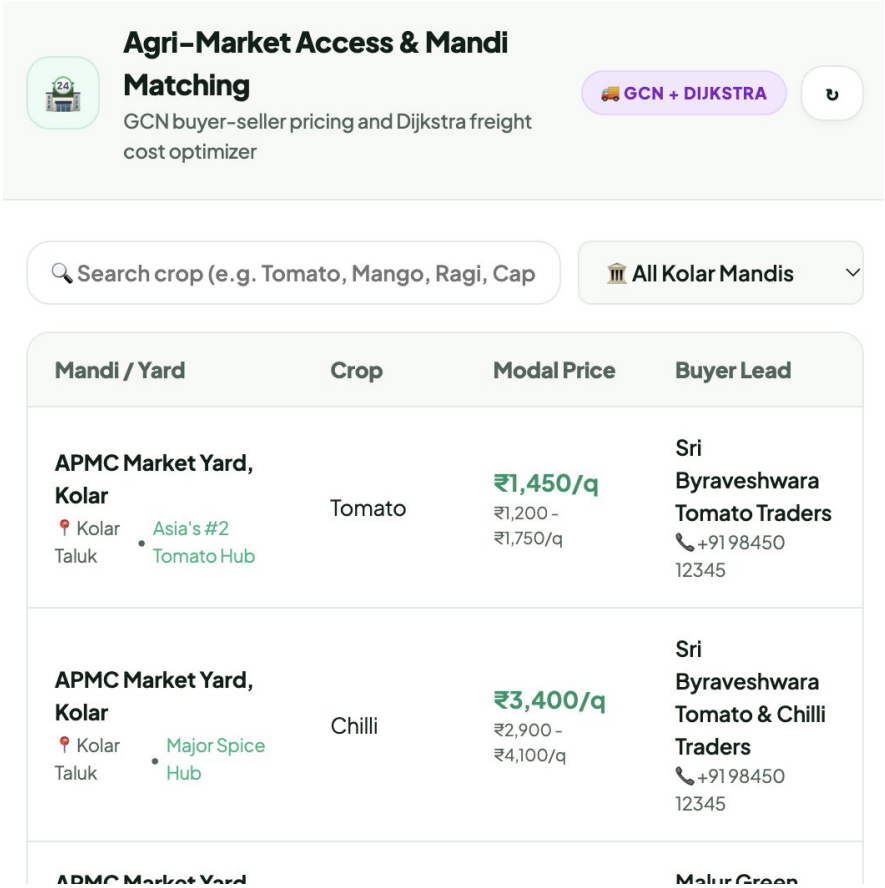


Fig 7.9: Mandi prices, buyer leads and freight optimisation

7.9 Voice Advisor and Community Model

Fig 7.10 shows the Multilingual Agricultural Voice Advisor and the Community Model. In the voice panel, the language is set to English, and Speak Now and Stop buttons control the microphone. A prompt line invites the user to tap Speak Now or select a chip: crop suitability status, pest alert and remedy, weather and rain forecast, mandi commodity prices or irrigation advice. In the community panel (FedAvg with $\varepsilon = 2.1$), four tiles show the model accuracy of

92.6 %, 284 edge clients, 12 federated rounds (converged) and the privacy budget ε = 2.1. A graph tracks accuracy between 90 % and 100 %, and a log at the bottom records events from the other modules, including the weather retrieval for Kolar, the tomato early blight diagnosis at 93.0 % confidence and 22.5 % leaf area, and the initialisation of the KrishiSense engine with its real on-device Naive Bayes and edge models.

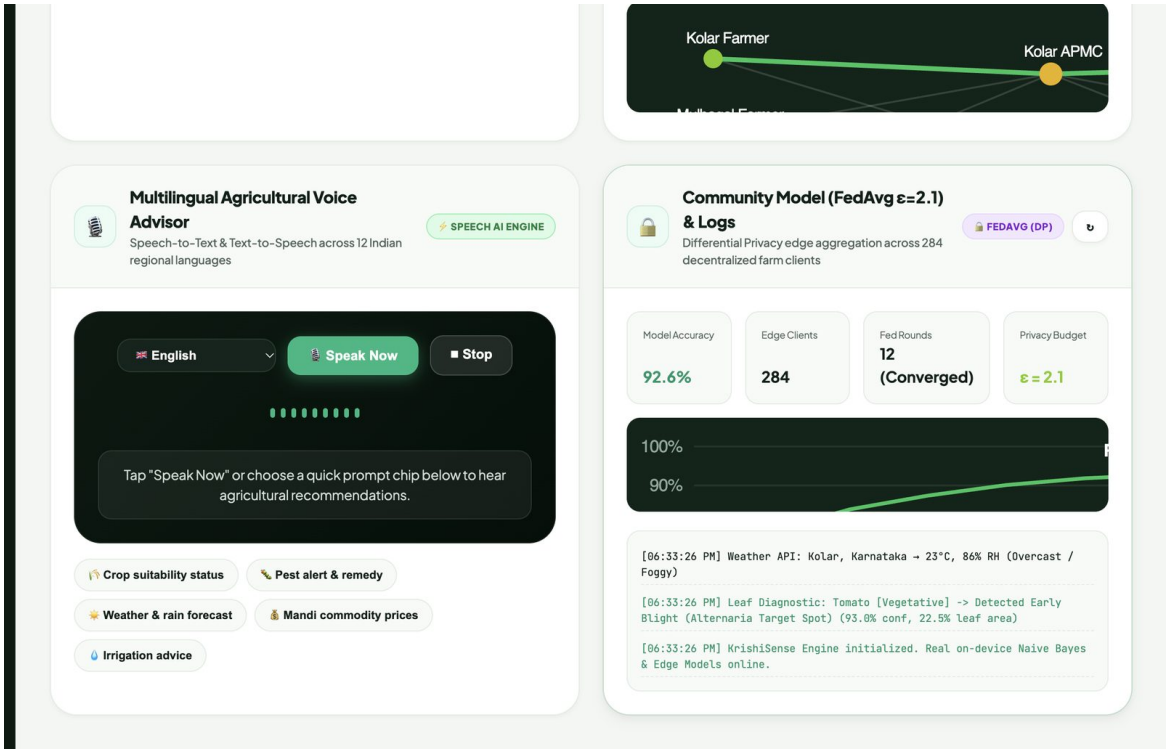


Fig 7.10: Multilingual voice advisor and community FedAvg model

7.10 Summary

The snapshots demonstrate the complete flow from weather and soil input, crop recommendation, a dated life-cycle plan, disease diagnosis, field sensing and market access to voice and community learning. The next chapter concludes the report and suggests directions for future work.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project set out to build an AI-driven decision-support system that guides a farmer through the entire crop life cycle and links the harvest to the market. The resulting platform, KrishiSense, brings together live weather, an on-device Gaussian Naive Bayes crop advisor trained on 2,200 records, a crop-specific growth and irrigation guide, a 19-step farm execution plan from Day 1 to Day 95, a leaf-disease and pest diagnostic lab, field sensor monitoring with imputation, a resource optimiser, a mandi and freight module based on graph methods and Dijkstra's algorithm, a multilingual voice advisor and a privacy-preserving community model.

The work demonstrates that the fragmented information a farmer normally needs can be unified in one interface and expressed as a dated, trackable sequence of actions. For the Kolar test case the system recommended tomato with 99 % suitability, generated a plan from 1 September to 4 December 2026, diagnosed early blight at 93.0 % confidence and displayed sustainability indicators of +33.5 % yield outlook, 72 % water conserved, 93 % pest-health index and 15 % logistics saving. The design is modular, runs largely on the device and is calibrated to a real district, so it can be extended to other regions and crops.

The project also produced a research paper, "AI for Rural Innovation and Sustainable Systems", published in IJSREM (Volume 10, Issue 10, October 2026), and gave the team practical experience in machine learning, computer vision, graph algorithms, federated learning and user-centred design. The limitations noted in Chapter 6 show that the indicators remain prototype outputs, and that validation in the field is the most important next step.

8.2 Expected Social and Environmental Impact

If adopted at scale, the system can reduce the knowledge gap between small farms and large agribusinesses. Precise irrigation reduces pressure on groundwater, which is a serious concern in Kolar. Early diagnosis and integrated pest management reduce the use of chemical pesticides and protect pollinators and beneficial insects. Better grading and routing reduce post-harvest losses, and direct contact with verified dealers improves the share of the final price that reaches the farmer. Because the platform is low in cost and available in regional languages and voice,

it can also support women farmers and first-time cultivators who have limited access to extension services.

8.2.1 Ethical and Responsible Use

Following the guidance of Omotayo et al. [9], the system is designed to be used responsibly, and the following principles apply to any future deployment.

- **Advisory role:** the output supports the decision of the farmer, an extension officer or an agricultural expert and does not replace it, especially in serious pest or disease outbreaks.
- **Transparency:** suitability percentages, confidence values and plain-language step guidance are always shown, so that the farmer can see how sure the model is.
- **Privacy and ownership:** soil values, field notes and plan status stay on the farmer's device, and community learning shares only clipped, noised updates.
- **Inclusion:** regional languages, voice access and a low-cost web platform help women farmers and first-time cultivators who have limited access to extension services.
- **Accountability:** buyer contacts should be those of verified APMC dealers, and prices should be confirmed in the market before produce is transported.

These principles, together with the field validation of Section 6.7.1, are the conditions under which the platform can move from a prototype to a dependable tool for sustainable rural development.

8.3 Future Work

The following directions are proposed for future work.

- **Field validation:** carry out multi-season trials on real farms, compare the recommendations, water use, yield and income with normal practice, and report MAE, RMSE, R^2, accuracy, precision, recall and F1-score.
- **Real IoT integration:** connect soil moisture, temperature and humidity sensors with wireless links and stream real-time data, as proposed in the paper.
- **Larger and wider datasets:** extend the crop model with more records, more crops and more districts, and add automatic recalibration for each region.
- **Yield forecasting:** add sequence models such as LSTM and GRU for yield prediction across growth stages using weather, NDVI and soil moisture histories.

- **Offline and low-bandwidth mode:** cache forecasts, prices and models, and deliver alerts through SMS and messaging applications.
- **More languages and dialects:** improve speech recognition for local dialects and add more regional languages and voice-first navigation.
- **Market integration:** connect directly to live APMC price feeds, e-marketplaces and transport providers, and add cold-chain and storage recommendations.
- **Explainability and trust:** show the factors behind each recommendation, allow expert review and record feedback from farmers.
- **Scalable deployment:** deploy the back end on a cloud platform with a database for history, user accounts for farmer groups and dashboards for extension officers.

8.4 Summary

The project achieved its objective of integrating soil testing, crop planning, protection, harvest and market access in one AI-driven platform. With field validation and real sensor integration, the framework can grow into a dependable tool for sustainable rural development.

CHAPTER 9

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Web Resources

[W1] Open-Meteo, Free Weather API, https://open-meteo.com

[W2] India Meteorological Department (IMD), https://mausam.imd.gov.in

[W3] European Space Agency, Sentinel-2 mission, https://sentinel.esa.int

[W4] NASA POWER Project, https://power.larc.nasa.gov

[W5] ISRIC, SoilGrids, https://soilgrids.org

[W6] Copernicus Climate Change Service, ERA5-Land reanalysis, https://climate.copernicus.eu

STATUS OF THE PUBLISHED RESEARCH PAPER

Publication Details

The work reported in this project was written up as a research paper and published in a peer-reviewed international journal. The details are given in Table A.1.

Table A.1: Details of the published paper

Item	Details
Title	AI for Rural Innovation and Sustainable Systems
Authors	Tanushree S R, Shreya A Hurakadli, Spoorthi M, Sneha K V and Dr. Arudra A
Affiliation	Department of Computer Science and Engineering, Rajiv Gandhi Institute of Technology, Bangalore, India
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USER MANUAL

B.1 Starting the Application

Open the KrishiSense index page in a modern browser on a computer, tablet or phone. Wait for the status indicator in the header to show Online, choose the language from the drop-down and, if desired, switch on Dark Mode for outdoor use.

B.2 Using the Main Functions

- **Check the weather:** open Live Weather (IMD), search for your village or select a quick region and read the temperature, rain probability and the 7-day forecast.
- **Choose a crop:** open Crop Advisor (AI), tap a Kolar crop preset or set the soil nutrient sliders, enter the planting date and the field soil-test values and press Generate Recommended Crop Plan.
- **Read the guide:** scroll to the crop guide to see the growth stages, water strategy and irrigation calendar.
- **Follow the plan:** use the Guided Farm Execution Plan; mark each step Done, In Progress or Pending and add field notes; filter by phase if required.
- **Scan a leaf:** open Leaf Disease and IoT Lab, select the crop and growth stage, upload a clear photograph of the affected leaf and read the diagnosis and remedy.
- **Check field conditions:** open Soil and Sensors to see moisture, pH, NDVI and other readings.
- **Plan the sale:** open Mandi and Supply Chain, search the crop, compare modal prices and contact the buyer lead; use the freight optimiser for the cheapest route.
- **Ask by voice:** open Voice Assistant, select the language, press Speak Now or tap a quick prompt and listen to the answer.

B.3 Precautions

- Enter soil values from a recent laboratory report or the Soil Health Card.
- Take leaf photographs in good light with the affected portion in focus.

- Treat the output as advice and consult an agricultural officer for serious pest or disease outbreaks.

DETAILED SCHEDULE OF THE EXECUTION PLAN

C.1 Nineteen-Step Plan for Tomato Planted on 1 September 2026

Table C.1 lists every step of the execution plan generated by the prototype for the Kolar test case, with its crop-day range, calendar dates and the key guidance displayed on the step card.

Table C.1: Detailed schedule of the 19 steps

No.	Step	Days	Dates (2026)	Key guidance
1	Soil Assessment and Health Baseline	1–3	1 Sep – 3 Sep	Sample top 15–20 cm soil; confirm N, P_2O_5, K_2O and pH (target 6.0–7.5)
2	Crop Selection and Variety Matching	4	4 Sep	Select certified high-yield variety suited to local climate and water; confirm sowing window with KVK
3	Land Preparation and Organic Tilth	5–10	5 Sep – 10 Sep	Ensure drainage furrows; install drip lines before sowing; 10–15 t/ha FYM or compost
4	Certified Seed Selection and Bio-Priming	11–12	11 Sep – 12 Sep	Seed germination above 85 %; Trichoderma viride at 4 g/kg and Rhizobium/PSB at 25 g/kg
5	Sowing / Transplanting Execution	13–15	13 Sep – 15 Sep	Field-capacity moisture before seeding; transplant sturdy 20–25 day seedlings in the evening

No.	Step	Days	Dates (2026)	Key guidance
6	Automated Irrigation and Moisture Management	16–95	16 Sep – 4 Dec	Stage-wise schedule from sensor telemetry and rain forecast; avoid excess irrigation before maturity
7	Split Fertilizer and Micronutrient Fertigation	22–62	22 Sep – 2 Nov	Drip fertigate with split potassium and calcium nitrate to prevent blossom end rot
8	Weed Control and Canopy Aeration	18–43	18 Sep – 14 Oct	Keep ridges weed-free so weeds do not harbour insect vectors and thrips
9	Field Telemetry and NDVI Canopy Vigour Auditing	18–86	18 Sep – 26 Nov	Target NDVI above 0.65 at peak vegetative flush; investigate local vigour drops
10	YOLOv8 Edge Pest Diagnostic Vision Lab	25–86	25 Sep – 26 Nov	Scout twice weekly; monitor Tuta absoluta and fruit borer with pheromone lures and Bt biologicals
11	Disease Early Warning and Micro-Climate Triggers	25–86	25 Sep – 26 Nov	Track humidity above 85 % and temperature spikes; apply Trichoderma or organic copper in cloudy spells
12	Integrated Pest Management Defence	25–86	25 Sep – 26 Nov	Maintain predators such as ladybird beetles and chrysoperla; spray only above ETL
13	Flowering and Reproductive Flush Audit	34–76	4 Oct – 16 Nov	Avoid chemical sprays between 8 AM and 11 AM during pollinator flight hours

No.	Step	Days	Dates (2026)	Key guidance
14	Yield Estimation and Harvest Scheduling	68–76	7 Nov – 16 Nov	Calibrate expected tonnage to organise crates, labour and vehicles
15	Pre-Harvest Readiness and Weather Scheduling	77–84	16 Nov – 24 Nov	Stop irrigation 5–8 days before harvest; clean crates and drying tarpaulins
16	Harvest Execution and Field Handling	85–95	24 Nov – 4 Dec	Harvest in cool hours; breaker stage for long haul, red ripe for local market
17	Post-Harvest Sorting, Grading and Cold Care	86–95	25 Nov – 4 Dec	Clean grading fetches 15–20 % higher modal price in Kolar APMC yards
18	Dijkstra Logistics Freight Optimization	87–95	26 Nov – 4 Dec	Stack ventilated crates or PICS hermetic bags securely to minimise transit loss
19	Mandi Market Access and Direct Dealer Connection	88–95	27 Nov – 4 Dec	Compare modal rates across Kolar, Srinivaspur, Bangarpet, Mulbagal and Malur; contact verified dealers